

CIMPA–ECCO 2026

Lecture Notes, Worksheets, and Solutions

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Lecture notes transcribed by Ignacio Rojas

August 2026

Please note that these notes were not provided or endorsed by the lecturer and have been significantly altered after the class. They may not accurately reflect the content covered in class and any errors are solely my responsibility.

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Part I

Lecture Notes

Chapter 1

Intro to Symmetric Functions and the Monomial Basis

Intro to Symmetric Functions and the monomial basis

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August 3, 2026

1.1 Why symmetric Functions?

- ◇ They encode polynomial Invariance: they are the functions that occur when the variables are unordered.
- ◇ They describe the roots of polynomials: impossibility of radical formulas.
- ◇ They provide a powerful enumeration tool.
- ◇ They describe representations of the symmetric and general linear groups.
- ◇ They connect to geometry: Schur functions represent Schubert classes in the cohomology of Grassmannians.
- ◇ They encode graphs: the chromatic symmetric function.

- ◇ They create a bridge between different disciplines: geometry, physics, probability, topology, graph theory, etc.
- ◇ They have applications to machine learning: in many data sets the order of the inputs does not matter.

1.2 What is a symmetric polynomial?

Question: What does a polynomial in variables x, y and z look like if we do not want it to change when we permute the variables?

$$f(x, y, z) = x^2 + y^2 + z^2$$

Does this polynomial change when we interchange x and y ? What about x and z ?

Example: $g(x, y, z) = x^2y + x^2z + y^2x + y^2z + z^2x + z^2y$ is “symmetric”.

Question: Can you give a concise description of this polynomial without writing all of the terms?

Example: $h(x, y, z) = x^2y + x^2z + y^2x + y^2z + z^2x + z^2y + 2x^3 + 2y^3 + 2z^3$ is also “symmetric”.

- ◇ We have seen that our polynomials will have several **variables**, we will use for any positive integer n

$$x_1, x_2, x_3, \dots, x_n$$

For our variables.

- ◇ The **coefficients** of the polynomials will come from $\mathbb{Q}$ (rational numbers, fractions).

$$g(x, y, z) = x_1^2x_2 + x_1^2x_3 + x_2^2x_1 + x_2^2x_3 + x_3^2x_1 + x_3^2x_2$$

- ◇ We need an “**action**” of the **symmetric group** to formalize the “interchange” of the variables.

1.3 Symmetric group and action on polynomials

- ◇ The **symmetric group** S_n is the group of all bijections from $[n] := \{1, 2, \dots, n\}$ to itself.

Example: When $n = 3$

$$S_3 = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\}$$

◇ The action of S_n on a polynomial with n variables, is defined for $\pi \in S_n$,

$$\pi \cdot f(x_1, x_2, \dots, x_n) = f(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(n)})$$

Example: $\pi = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ and $f(x_1, x_2, x_3) = x_1^2 x_2^2 x_3 + x_1^2 x_3^2 x_2 + x_2^2 x_3^2 x_1$

Then

$$\begin{aligned} \pi \cdot f(x_1, x_2, x_3) &= x_{\pi(1)}^2 x_{\pi(2)}^2 x_{\pi(3)} + x_{\pi(1)}^2 x_{\pi(3)}^2 x_{\pi(2)} + x_{\pi(2)}^2 x_{\pi(3)}^2 x_{\pi(1)} \\ &= x_3^2 x_1^2 x_2 + x_3^2 x_2^2 x_1 + x_1^2 x_2^2 x_3 \end{aligned}$$

NOTE: $\pi \cdot f(x_1, x_2, x_3) = f(x_1, x_2, x_3)$

Example: $\pi = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ and $f(x_1, x_2, x_3, x_4) = x_1^2 + 3x_2x_4 - 2x_3^2x_4$

Then

$$\pi \cdot f(x_1, x_2, x_3, x_4) = x_{\pi(1)}^2 + 3x_{\pi(2)}x_{\pi(4)} - 2x_{\pi(3)}^2x_{\pi(4)} = x_2^2 + 3x_4x_3 - 2x_1^2x_3$$

NOTE: $\pi \cdot f(x_1, x_2, x_3, x_4) \neq f(x_1, x_2, x_3, x_4)$

Proposition 1.3.1 (Properties of the action). Let $c \in \mathbb{Q}$, f and g polynomials and $\sigma, \pi \in S_n$. Then

(i) $\pi \cdot (cf) = c(\pi \cdot f)$

(ii) $\pi \cdot (f + g) = \pi \cdot f + \pi \cdot g$

(iii) $\pi \cdot (fg) = (\pi \cdot f)(\pi \cdot g)$

(iv) $(\pi\sigma) \cdot f = \pi \cdot (\sigma \cdot f)$

1.4 Definition of symmetric polynomial

Definition 1.4.1. A polynomial $f(x_1, x_2, \dots, x_n)$ is called **symmetric** if $\pi \cdot f(x_1, \dots, x_n) = f(x_1, \dots, x_n)$ for all $\pi \in S_n$.

We use $\Lambda(x_1, \dots, x_n)$ or Λ_n for the set of all symmetric polynomials in n variables.

Example: $x_1^2x_2 + x_1^2x_3 + x_2^2x_1 + x_2^2x_3 + x_3^2x_1 + x_3^2x_2 \in \Lambda_3$.

◇ Since we can add and scale symmetric polynomials and get a symmetric polynomial, Λ_n is an infinite dimensional vector space.

Definition 1.4.2. A polynomial $f(x_1, x_2, \dots, x_n)$ is **homogeneous of degree k** if every one of its monomials has degree k .

- ◇ We use Λ_n^k for the set of all homogenous symmetric polynomials of degree k .
- ◇ **Note:** Every monomial in the 0 polynomial has degree k (since it has no monomials), so $0 \in \Lambda_n^k$.
- ◇ Λ_n can be decomposed into its homogeneous pieces:

$$\Lambda_n = \Lambda_n^0 \oplus \Lambda_n^1 \oplus \Lambda_n^2 \oplus \dots = \bigoplus_k \Lambda_n^k \quad \text{Note: } \Lambda_n^0 = \mathbb{Q}.$$

- ◇ What does this say? If $f \in \Lambda_n$, then $f = f_0 + f_1 + f_2 + \dots$, where $f_i \in \Lambda_n^i$.
- ◇ Λ_n^k is finite dimensional, and we concentrate on understanding these homogeneous spaces.

1.5 Understanding Λ_n^k

The first step to understand a vector space is to find a linear basis for it.

Question: What is the symmetric polynomial with the smallest number of monomials that contains $x_1^2 x_2$ in Λ_3^3 ?

Question: What is the symmetric polynomial with the smallest number of monomials that contains x_1^3 in Λ_3^3 ?

Question: What is the symmetric polynomial with the smallest number of monomials that contains $x_1 x_2 x_3$ in Λ_3^3 ?

Question: How can we describe these polynomials without listing all the monomials?

Algebra Note: We can think of the symmetric group S_n acting on the set of all monomials: For $n = 3$ and $k = 3$ the set is $\{x_1 x_2 x_3, x_1^2 x_2, x_1^2 x_3, x_2^2 x_1, x_2^2 x_3, x_3^2 x_1, x_3^2 x_2, x_1^3, x_2^3, x_3^3\}$

Question: What are the orbits of the action?

$$\begin{aligned} O_1 &= \{x_1 x_2 x_3\} \\ O_2 &= \{x_1^2 x_2, x_1^2 x_3, x_2^2 x_1, x_2^2 x_3, x_3^2 x_1, x_3^2 x_2\} \\ O_3 &= \{x_1^3, x_2^3, x_3^3\} \end{aligned}$$

Question: How can we “index” these orbits, what is their defining property? **PARTITIONS!**

Note: the polynomials we got above are the sum of the elements in the orbits. The partitions of $k = 3$ are $(1, 1, 1)$, $(2, 1)$ and (3)

$$m_{1,1,1} = x_1 x_2 x_3$$

$$m_{2,1} = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_1 + x_2^2 x_3 + x_3^2 x_1 + x_3^2 x_2$$

$$m_3 = x_1^3 + x_2^3 + x_3^3$$

1.6 Partitions

Definition 1.6.1. For any nonnegative integer k , a **partition** λ of k is a weakly decreasing sequence of nonnegative integers $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$ such that $\lambda_1 + \lambda_2 + \dots + \lambda_\ell = k$.

Example: The partitions of $k = 5$ are (5) , $(4, 1)$, $(3, 2)$, $(3, 1, 1)$, $(2, 2, 1)$, $(2, 1, 1, 1)$, and $(1, 1, 1, 1, 1)$

If λ is a partition of k , we write $\lambda \vdash k$ or $|\lambda| = \lambda_1 + \lambda_2 + \dots + \lambda_\ell = k$.

The λ_i are called **parts**, and the number of nonzero parts is called the **length** of λ and we write $\ell(\lambda)$.

We use exponent notation for partitions: $(k^{m_k}, (k-1)^{m_{k-1}}, \dots, 2^{m_2}, 1^{m_1})$

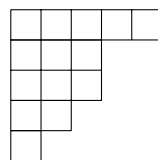
For example, $(5, 5, 5, 4, 3, 3, 3, 3, 1, 1, 1, 1) = (5^3, 4^1, 3^4, 1^4)$.

We call the m_i **multiplicities**, and this means there are m_i parts i in λ .

For example in the partition above, 5 has multiplicity 3, so $m_5 = 3$.

The **Young diagram**: It is useful to associate a geometric picture to a partition.

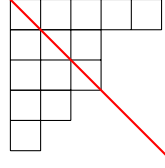
For example, $(5, 3, 3, 2, 1)$ has Young diagram:



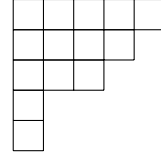
1.7 Partitions (continued)

- ◇ The **conjugate** of a partition λ is the partition λ' with parts the column lengths of the Young diagram of λ .

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$$\lambda = (5, 3, 3, 2, 1)$$



$$\lambda' = (5, 4, 3, 1, 1)$$

- ◇ The conjugate map is an involution, $(\lambda')' = \lambda$. It is a bijection!
- ◇ The number of partitions will be the number $p(k)$.

k	0	1	2	3	4	5	6	7	8
$p(k)$	1	1	2	3	5	7	11	15	22

- ◇ The generating function for partitions:

Euler's Theorem:

$$\sum_{k=0}^{\infty} p(k)x^k = \prod_{i=1}^{\infty} \frac{1}{1-x^i}$$

1.8 The monomial symmetric polynomials

Definition 1.8.1. For $n \geq 1$ and a partition λ , the **monomial symmetric polynomial** $m_{\lambda}(x_1, \dots, x_n)$ indexed by λ is the sum of the monomial $x_1^{\lambda_1} x_2^{\lambda_2} \dots x_{\ell(\lambda)}^{\lambda_{\ell(\lambda)}}$ and all the possible **different** monomials $\pi \cdot (x_1^{\lambda_1} x_2^{\lambda_2} \dots x_{\ell(\lambda)}^{\lambda_{\ell(\lambda)}})$ obtained by acting by $\pi \in S_n$.

Note: If $\ell(\lambda) > n$, then $m_{\lambda}(x_1, \dots, x_n) = 0$ because we assume that $x_j = 0$ for $j > n$.

Examples:

$$m_{2,1}(x_1, x_2) = x_1^2 x_2 + x_1 x_2^2$$

$$m_{2,1}(x_1, x_2, x_3) = x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_3^2 + x_2 x_3^2$$

$$m_{3,3,1,1}(x_1, x_2, x_3, x_4) = x_1^3 x_2^3 x_3 x_4 + x_1^3 x_2 x_3^3 x_4 + x_1^3 x_2 x_3 x_4^3 + x_1 x_2^3 x_3^3 x_4 + x_1 x_2^3 x_3 x_4^3 + x_1 x_2 x_3^3 x_4^3$$

$$m_k(x_1, x_2, x_3, x_4) = x_1^k + x_2^k + x_3^k + x_4^k$$

$$m_{1,1,1}(x_1, x_2, x_3, x_4) = x_1 x_2 x_3 + x_1 x_2 x_4 + x_1 x_3 x_4 + x_2 x_3 x_4$$

1.9 The monomial symmetric functions are symmetric

Proposition 1.9.1. *For all positive integers n and all partitions λ , the polynomial $m_\lambda(x_1, \dots, x_n)$ is a symmetric polynomial in $x_1, x_2, \dots, x_n$.*

Proof. If $n < \ell(\lambda)$, then $m_\lambda(x_1, \dots, x_n) = 0$, which is symmetric.

Otherwise, let $\sigma_i = (i, i+1)$ be the permutation in S_n that maps i to $i+1$ and $i+1$ to i and maps all other $j \neq i, i+1$ to j ($j \rightarrow j$).

Claim: $\sigma_i \cdot m_\lambda(x_1, \dots, x_n) = m_\lambda(x_1, \dots, x_n)$ for any $1 \leq i \leq n-1$.

To show this we must show that every monomial $x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$ in $\sigma_i \cdot m_\lambda(x_1, \dots, x_n)$ has the same coefficient in both $\sigma_i \cdot m_\lambda(x_1, \dots, x_n)$ and $m_\lambda(x_1, \dots, x_n)$.

Note that the coefficient of $x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$ is 1 in $m_\lambda(x_1, \dots, x_n)$ if $(\alpha_1, \alpha_2, \dots, \alpha_n)$ is a reordering of λ (we could add some zeros at the end of λ as some α_j 's could be zero) and the coefficient is zero otherwise.

On the other hand, the coefficient of $x_1^{\alpha_1} \dots x_i^{\alpha_{i+1}} x_{i+1}^{\alpha_i} \dots x_n^{\alpha_n}$ in $\sigma_i \cdot m_\lambda(x_1, \dots, x_n)$ is the same as the coefficient of $x_1^{\alpha_1} \dots x_i^{\alpha_i} x_{i+1}^{\alpha_{i+1}} \dots x_n^{\alpha_n}$ in $m_\lambda(x_1, \dots, x_n)$ since permutation does not change the coefficients. And the coefficient is 1 if $(\alpha_1, \alpha_2, \dots, \alpha_n)$ is a reordering of λ and zero otherwise.

But $(\alpha_1, \alpha_2, \dots, \alpha_n)$ is a reordering of λ if and only if $(\alpha_1, \alpha_2, \dots, \alpha_{i+1}, \alpha_i, \dots, \alpha_n)$ is also a reordering of λ . Therefore, the claim follows. $\square$

1.10 The monomial symmetric polynomials form a basis for Λ_n^k

Proposition 1.10.1. *If $n \geq 1$, $k \geq 0$, and $n \geq k$, then $\{m_\lambda(x_1, \dots, x_n) \mid \lambda \vdash k\}$ is a basis for Λ_n^k . In particular, $\dim(\Lambda_n^k) = p(k)$, the number of partitions of k .*

Proof. Note that if $\lambda \neq \mu$, then $m_\lambda(x_1, \dots, x_n)$ and $m_\mu(x_1, \dots, x_n)$ have no terms in common. And $m_\lambda(x_1, \dots, x_n) \neq 0$ for all $\lambda \vdash k$ because $n \geq k$. Then

$$\sum_{\lambda \vdash k} a_\lambda m_\lambda(x_1, \dots, x_n) = 0$$

if and only if $a_\lambda = 0$ for all $\lambda \vdash k$. Therefore, $\{m_\lambda(x_1, \dots, x_n) \mid \lambda \vdash k\}$ is linearly independent.

Now we show that $\{m_\lambda(x_1, \dots, x_n) \mid \lambda \vdash k\}$ spans Λ_n^k . Let $f \in \Lambda_n^k$, if $f = 0$, we are done. So assume $f \neq 0$. We use induction on the number of monomials in f to show that $f = \sum_\lambda c_\lambda m_\lambda(x_1, \dots, x_n)$.

Since $f \neq 0$, it has a term $cx_1^{\alpha_1} \dots x_n^{\alpha_n}$ with $c \in \mathbb{Q}$ and not equal to zero and $\alpha \vdash k$. Since f is symmetric, then for every $\pi \in S_n$, $\pi \cdot (cx_1^{\alpha_1} \dots x_n^{\alpha_n}) = c\pi \cdot (x_1^{\alpha_1} \dots x_n^{\alpha_n})$ is also a term in f . Therefore, $f - cm_\alpha \in \Lambda_n^k$ and has fewer terms than f . By induction $f - cm_\alpha$ is a linear combination of the $m_\lambda(x_1, \dots, x_n)$'s. Thus $\{m_\lambda(x_1, \dots, x_n) \mid \lambda \vdash k\}$ spans Λ_n^k .

Hence, $\{m_\lambda(x_1, \dots, x_n) \mid \lambda \vdash k\}$ is a basis and therefore the dimension is the $p(k)$. $\square$

1.11 Symmetric Functions

Note: If $n \geq k$, then $\dim(\Lambda_n^k) = p(k)$ and it is independent of n .

If we let $n \rightarrow \infty$, the polynomials $m_\lambda(x_1, x_2, \dots, x_n)$ become power series.

$$\begin{aligned} m_{2,1}(x_1, x_2) &= x_1^2 x_2 + x_1 x_2^2 \\ m_{2,1}(x_1, x_2, x_3) &= x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_3^2 + x_2 x_3^2 \\ m_{2,1}(x_1, x_2, x_3, x_4) &= x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_3^2 + x_2 x_3^2 \\ &\quad + x_1^2 x_4 + x_2^2 x_4 + x_3^2 x_4 + x_1 x_4^2 + x_2 x_4^2 + x_3 x_4^2 \end{aligned}$$

$$m_{2,1} = m_{2,1}(x_1, x_2, \dots) = \sum_{1 \leq i \neq j \leq \infty} x_i^2 x_j$$

A power series is like an infinite polynomial. And it is called a **function** if it has bounded degree, that is the sum of the exponents is finite for every monomial.

Hence, a **symmetric function** is a power series with bounded degree such that it doesn't change when we permute the variables:

$$f = \sum c_\alpha x^\alpha \quad (\text{here } x^\alpha \text{ is short for } x_1^{\alpha_1} x_2^{\alpha_2} \dots)$$

$\alpha = (\alpha_1, \alpha_2, \dots)$ is a sequence of nonnegative integer and $\alpha_1 + \alpha_2 + \dots$ is bounded and $\pi \cdot f = f$.

Proposition 1.11.1. For any λ , if $n \geq \ell(\lambda)$, then $m_\lambda(x_1, x_2, \dots, x_n, 0, \dots, 0) = m_\lambda(x_1, \dots, x_n)$.

Proof. We can see that $m_\lambda(x_1, x_2, \dots, x_n, 0) = m_\lambda(x_1, \dots, x_n)$ because they are the sum of the same monomials, and the general result follows by induction. $\square$

1.12 Multiplying power series

Given $F(x) = \sum_{i=0}^{\infty} a_i x^i$ and $G(x) = \sum_{i=0}^{\infty} b_i x^i$, then $F(x)G(x) = \sum_{k=0}^{\infty} c_k x^k$ where $c_k = \sum_{i=0}^k a_i b_{k-i}$

Geometric Series: $\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$

Example: What is $(\sum_{i=0}^{\infty} x^i) (\sum_{i=0}^{\infty} x^i)$?

$$\left(\sum_{i=0}^{\infty} x^i \right) \left(\sum_{i=0}^{\infty} x^i \right) = (1 + x + x^2 + x^3 + \dots)(1 + x + x^2 + x^3 + \dots) = 1 + 2x + 3x^2 + 4x^3 + \dots$$

The coefficient is: $c_k = \sum_{i=0}^k (1)(1) = k + 1$

What about multivariate power series?

Example: What is $m_1 m_1$?

$$m_1 m_1 = (x_1 + x_2 + x_3 + \dots)(x_1 + x_2 + x_3 + \dots)$$

We get quadratic polynomials: x_i^2 and $x_i x_j$ with $i \neq j$ what are their coefficients?

To get x_1^2 we must pick x_1 from both factors, so coefficient is 1. We can get $x_1 x_2$ in two ways pick x_1 from either first or second factor (two choices) and the x_2 from the other factor.

$$m_1 m_1 = m_2 + 2m_{1,1}$$

1.13 Multiplying monomial symmetric functions

n	$m_{1,1}(x_1, \dots, x_n) m_{2,1}(x_1, \dots, x_n)$
2	$m_{3,2}$
3	$m_{3,2} + 2m_{3,1,1} + 2m_{2,2,1}$
4	$m_{3,2} + 2m_{3,1,1} + 2m_{2,2,1} + 3m_{2,1,1,1}$
5	$m_{3,2} + 2m_{3,1,1} + 2m_{2,2,1} + 3m_{2,1,1,1}$
6	$m_{3,2} + 2m_{3,1,1} + 2m_{2,2,1} + 3m_{2,1,1,1}$
7	$m_{3,2} + 2m_{3,1,1} + 2m_{2,2,1} + 3m_{2,1,1,1}$

Note: the product stabilizes when you have “enough” variables and stays the same.

It will be enough to compute the coefficient of $x_1^{\alpha_1} \dots x_{\alpha_\ell}^{\alpha_\ell}$ because the product is symmetric.

Proposition 1.13.1. For any partitions λ, μ and $\nu \vdash |\lambda| + |\mu|$, the coefficients $a_{\lambda, \mu}^\nu$ in the

product

$$m_\lambda(x_1, \dots, x_n) m_\mu(x_1, \dots, x_n) = \sum_{\nu \vdash |\lambda| + |\mu|} a_{\lambda, \mu}^\nu m_\nu(x_1, \dots, x_n)$$

do not depend on n when $n \geq \ell(\lambda) + \ell(\mu)$ and

$$m_\lambda m_\mu = \sum_{\nu \vdash |\lambda| + |\mu|} a_{\lambda, \mu}^\nu m_\nu$$

i.e., for the functions the coefficients are the same.

1.14 The monomial symmetric functions form a basis

We use Λ for the ring of symmetric functions.

And Λ^k for the homogeneous symmetric functions of degree k .

Proposition 1.14.1. *For all $k \geq 0$, $\{m_\lambda \mid \lambda \vdash k\}$ is a basis for Λ^k . And Λ^k has dimension $p(k)$.*

Proof. Linear independence is the same as for polynomials. For $\lambda \neq \mu$, m_λ and m_μ have all different monomials. So, any linear combination of the m_λ that results in 0 must have all coefficients equal to zero.

For the spanning property, we proceed again by induction, but now the induction is on the number of different partitions that can occur as powers in a monomial. Since, there are only finite number of partitions $p(k)$, we will have a finite sum of the m_λ 's. Given a function $f \in \Lambda^k$. Suppose that there is monomial $cx_1^{\lambda_1} x_2^{\lambda_2} \dots x_{\ell(\lambda)}^{\lambda_{\ell(\lambda)}}$ with λ a partition and $c \in \mathbb{Q}$. Since f is symmetric, every permutation of this monomial also occurs with the same coefficient. So, cm_λ is a summand of f . If λ is the only partition occurring as powers of a monomial, then $f = cm_\lambda$. Otherwise, $f - cm_\lambda \in \Lambda^k$ and nonzero and by induction, it has fewer partitions as powers of a monomial. By induction, $f - cm_\lambda$ is the span of $\{m_\lambda \mid \lambda \vdash k\}$ and so is f . $\square$

Chapter 2

Elementary, Homogeneous, and Power Bases

Elementary, homogeneous, and power bases

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August 4, 2026

2.1 The elementary symmetric functions – Motivation

Consider expanding the polynomial:

$$f(z) = (z - x_1)(z - x_2) \dots (z - x_n) = z^n + c_{n-1}z^{n-1} + c_{n-2}z^{n-2} + \dots + c_1z + c_0$$

Note that the coefficients $c_i(x_1, \dots, x_n)$ are given by

$$c_i(x_1, \dots, x_n) = (-1)^{n-i} m_{\underbrace{1, \dots, 1}_{n-i}}(x_1, \dots, x_n)$$

Examples:

◇ For $n = 3$:

$$\begin{aligned} (z - x_1)(z - x_2)(z - x_3) &= z^3 - (x_1 + x_2 + x_3)z^2 + (x_1x_2 + x_1x_3 + x_2x_3)z - x_1x_2x_3 \\ &= z^3 - m_1(x_1, x_2, x_3)z^2 + m_{1,1}(x_1, x_2, x_3)z - m_{1,1,1}(x_1, x_2, x_3) \end{aligned}$$

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◇ For $n = 4$:

$$(z - x_1)(z - x_2)(z - x_3)(z - x_4) = z^4 - m_1 z^3 + m_{1,1} z^2 - m_{1,1,1} z + m_{1,1,1,1}$$

2.2 Definition of elementary symmetric function

Definition 2.2.1. For any integer $k \geq 1$, the k -th elementary symmetric function e_k is defined as

$$e_k = m_{1^k} = \sum_{1 \leq i_1 < i_2 < \dots < i_k} x_{i_1} x_{i_2} \dots x_{i_k}$$

We define $e_0 := 1$, and for $k < 0$, $e_k := 0$.

For any partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$, we define

$$e_\lambda = e_{\lambda_1} e_{\lambda_2} \dots e_{\lambda_\ell}$$

Examples:

$$e_3 = \sum_{1 \leq i < j < k} x_i x_j x_k = m_{1,1,1}$$

$$\begin{aligned} e_{2,1} &= e_2 e_1 = \left(\sum_{1 \leq i < j} x_i x_j \right) \left(\sum_{k \geq 1} x_k \right) = (x_1 x_2 + x_1 x_3 + x_2 x_3 + \dots)(x_1 + x_2 + x_3 + \dots) \\ &= 3x_1 x_2 x_3 + \dots + x_1^2 x_2 + x_1 x_2^2 + \dots = 3m_{1,1,1} + m_{2,1} \end{aligned}$$

Fact: $e_\lambda = \sum_{\mu} M_{\lambda,\mu}(e, m) m_\mu$, where $M_{\lambda,\mu}(e, m)$ is a non-negative integer.

2.3 A combinatorial description for $M_{\lambda,\mu}(e, m)$

To describe the coefficients $M_{\lambda,\mu}(e, m)$, we fill the boxes of the Young diagram of shape λ with positive integers.

- ◇ A tableau is **row strict** if the entries strictly increase from left to right in every row.
- ◇ The **content** μ of a tableau is the sequence $\mu = (\mu_1, \mu_2, \dots)$ where μ_i is the number of times i appears in the tableau.

Proposition 2.3.1. For partitions λ and μ , $M_{\lambda,\mu}(e, m)$ is the number of row strict tableaux of shape λ and content μ .

Proof. Recall $e_\lambda = e_{\lambda_1} e_{\lambda_2} \dots e_{\lambda_{\ell(\lambda)}}$. A monomial in e_λ is obtained by choosing one monomial $x_{j_1} x_{j_2} \dots x_{j_{\lambda_i}}$ with $j_1 < j_2 < \dots < j_{\lambda_i}$ from each factor e_{λ_i} . Place these indices $j_1, j_2, \dots, j_{\lambda_i}$ in the i -th row of the Young diagram of λ . Since $j_1 < j_2 < \dots < j_{\lambda_i}$, the rows are strictly increasing.

To get a monomial corresponding to m_μ , the integer i must be chosen μ_i times in total across all rows, so the content of the tableau is μ . Thus, the coefficient $M_{\lambda,\mu}(e, m)$ is precisely the number of such row strict tableaux. $\square$

2.4 Worked Example: $e_{4,2,1}$ in the monomial basis

Example: Write $e_{4,2,1}$ in the monomial basis.

Here the shape is $\lambda = (4, 2, 1)$.

- ◇ Row 1 has length 4, so it requires 4 distinct entries (strictly increasing).
- ◇ Row 2 has length 2, requiring 2 distinct entries.
- ◇ Row 3 has length 1, requiring 1 entry.

Since each entry i can appear at most once per row, and there are $\ell(\lambda) = 3$ rows, no entry i can appear more than 3 times. Thus $\mu_i \leq 3$ for all i .

The non-zero coefficients $M_{(4,2,1),\mu}(e, m)$ correspond to row-strict tableaux of shape $(4, 2, 1)$ with content μ . Counting these tableaux yields the expansion:

$$e_{4,2,1} = m_{4,2,1} + m_{3,3,1} + 2m_{3,2,2} + 2m_{3,2,1,1} + m_{3,1,1,1,1} + 4m_{2,2,2,1} + 4m_{2,2,1,1,1} + 10m_{2,1,1,1,1,1} + 30m_{1^7}$$

2.5 Lexicographic order and the transition matrix

Definition 2.5.1. The **lexicographic order** $>_{\text{lex}}$ on partitions of k is defined by: $\lambda >_{\text{lex}} \mu$ if the first non-zero entry in the sequence $\lambda - \mu = (\lambda_1 - \mu_1, \lambda_2 - \mu_2, \dots)$ is positive.

For $k = 3$, the partitions ordered lexicographically are $(3) >_{\text{lex}} (2, 1) >_{\text{lex}} (1, 1, 1)$.

The matrix $[M_{\lambda,\mu}(e, m)]$ for $k = 3$, with rows indexed by $\lambda \in \{(1^3), (2, 1), (3)\}$ and columns by $\mu \in \{(1^3), (2, 1), (3)\}$, is given by:

$$\begin{pmatrix} (1^3) \\ (2, 1) \\ (3) \end{pmatrix} \begin{pmatrix} (1^3) & (2, 1) & (3) \end{pmatrix} \begin{bmatrix} 6 & 3 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

2.6 The matrix $[M_{\lambda,\mu}(e, m)]$ is triangular

Proposition 2.6.1. *Let $\lambda, \mu \vdash k$. The following statements hold:*

- (i) *If $\mu >_{\text{lex}} \lambda'$, then $M_{\lambda,\mu}(e, m) = 0$.*
- (ii) *$M_{\lambda,\lambda'}(e, m) = 1$.*

Proof. (i) and (ii) For $M_{\lambda,\mu}(e, m)$ to be non-zero, there must exist a row strict tableau of shape λ and content μ . Since the rows are strictly increasing, each entry i can appear at most once in any given row. Therefore, entry 1 can appear in at most $\ell(\lambda) = \lambda'_1$ rows, so $\mu_1 \leq \lambda'_1$.

If $\mu_1 > \lambda'_1$, no such tableau can exist, so $M_{\lambda,\mu}(e, m) = 0$. If $\mu_1 = \lambda'_1$, then 1 must appear in every row (i.e. in the first column). Proceeding by induction on content, if $\mu >_{\text{lex}} \lambda'$, we obtain $M_{\lambda,\mu}(e, m) = 0$.

When $\mu = \lambda'$, there is a unique choice: entry j must fill every box in the j -th column, so row i contains the entries $1, 2, \dots, \lambda_i$. This gives a unique row strict tableau of shape λ and content λ' , so $M_{\lambda,\lambda'}(e, m) = 1$. $\square$

2.7 The elementary symmetric functions form a basis

Proposition 2.7.1. *The set $\{e_\lambda \mid \lambda \vdash k\}$ is a basis for Λ^k , called the *elementary symmetric basis*.*

Proof. The change of basis matrix $A = [M_{\lambda,\mu'}(e, m)]$ expressing $\{e_\lambda\}$ in terms of $\{m_\mu\}$ is unitriangular (lower triangular with 1's on the diagonal) when rows are ordered by λ and columns by μ' in lexicographic order. Since $\det(A) = 1 \neq 0$, the matrix A is invertible, and hence $\{e_\lambda \mid \lambda \vdash k\}$ is a basis for Λ^k . $\square$

Example: For $k = 4$, the change of basis matrix A is:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 & 0 \\ 6 & 2 & 1 & 0 & 0 \\ 12 & 5 & 2 & 1 & 0 \\ 24 & 12 & 6 & 4 & 1 \end{bmatrix}$$

Fundamental Theorem of Symmetric Functions: Every symmetric function can be uniquely expressed as a polynomial in the elementary symmetric functions $e_1, e_2, e_3, \dots$. That is, $\Lambda = \mathbb{Q}[e_1, e_2, e_3, \dots]$.

Generating Function for e_k :

$$E(t) = \sum_{k=0}^{\infty} e_k t^k = \prod_{i=1}^{\infty} (1 + x_i t)$$

2.8 The complete homogeneous symmetric functions

Definition 2.8.1. For any integer $k \geq 1$, the **k -th complete homogeneous symmetric function h_k** is defined as the sum of all monomials of degree k :

$$h_k = \sum_{1 \leq i_1 \leq i_2 \leq \dots \leq i_k} x_{i_1} x_{i_2} \dots x_{i_k}$$

We define $h_0 := 1$, and for $k < 0$, $h_k := 0$.

For any partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$, we define

$$h_\lambda = h_{\lambda_1} h_{\lambda_2} \dots h_{\lambda_\ell}$$

Examples:

$$h_3 = \sum_{1 \leq i \leq j \leq k} x_i x_j x_k = m_3 + m_{2,1} + m_{1,1,1}$$

$$h_{2,1} = h_2 h_1 = (m_2 + m_{1,1}) m_1 = m_3 + 2m_{2,1} + 3m_{1,1,1}$$

Fact: $h_k = \sum_{\lambda \vdash k} m_\lambda$.

2.9 The coefficients $M_{\lambda,\mu}(h, m)$

For $k = 3$, the transition matrix $[M_{\lambda,\mu}(h, m)]$ with rows $\lambda \in \{(1^3), (2, 1), (3)\}$ and columns $\mu \in \{(1^3), (2, 1), (3)\}$ is:

$$\begin{bmatrix} 6 & 3 & 1 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Proposition 2.9.1. $M_{\lambda,\mu}(h, m)$ is equal to the number of tableaux of shape λ with weakly increasing rows and content μ .

Example: For $h_{4,2,1}$, the matrix is non-triangular, symmetric, and has determinant 1.

2.10 The generating function for h_k

Proposition 2.10.1. The generating function for h_k is

$$H(t) = \sum_{k \geq 0} h_k t^k = \prod_{i=1}^{\infty} \frac{1}{1 - x_i t}$$

Proof. Note that $\frac{1}{1-x_i t} = 1 + x_i t + x_i^2 t^2 + \dots$. Each summand indicates the power of x_i in the monomial. When we multiply $(1 + x_1 t + x_1^2 t^2 + \dots)(1 + x_2 t + x_2^2 t^2 + \dots) \dots$ we generate every possible monomial, and the coefficient of t^k is the sum of all possible monomials of degree k . This is precisely what h_k is. $\square$

Proposition 2.10.2. For all $k \geq 1$,

$$\sum_{j \geq 0} (-1)^j e_j h_{k-j} = 0$$

Proof. Recall $E(t) = \sum_{k \geq 0} e_k t^k = \prod_{i=1}^{\infty} (1 + x_i t)$, hence $E(-t) = \sum_{k \geq 0} (-1)^k e_k t^k = \prod_{i=1}^{\infty} (1 - x_i t)$. Thus,

$$E(-t)H(t) = 1 = (e_0 - e_1 t + e_2 t^2 - \dots)(h_0 + h_1 t + h_2 t^2 + \dots)$$

We use the formula for the coefficient in the product of two power series:

$$c_k = \sum_{j=0}^k a_j b_{k-j} \quad \text{where } a_j = (-1)^j e_j \text{ and } b_{k-j} = h_{k-j}$$

The result follows. $\square$

2.11 The complete homogeneous symmetric functions form a basis

Proposition 2.11.1. $\{h_\lambda \mid \lambda \vdash k\}$ is a basis for Λ^k , called the *complete homogeneous symmetric basis*.

Proof. Since $\{h_\lambda \mid \lambda \vdash k\}$ contains $p(k)$ symmetric functions, it is sufficient to show that it spans Λ^k . We prove this by induction on k .

For $k = 0$, $h_0 = e_0 = 1$ and $\Lambda^0 = \text{Span}\{1\} = \text{Span}\{h_0\}$. For $k = 1$, $h_1 = e_1 = x_1 + x_2 + \dots$ and $\Lambda^1 = \text{Span}\{e_1\} = \text{Span}\{h_1\}$.

Now assume $k \geq 2$. We have $\text{Span}\{h_\lambda \mid \lambda \vdash k\} \subseteq \Lambda^k$ because the h_λ are symmetric and homogeneous of degree k .

To show $\Lambda^k \subseteq \text{Span}\{h_\lambda \mid \lambda \vdash k\}$, since $\{e_\lambda \mid \lambda \vdash k\}$ is a basis, it suffices to show that $e_\mu \in \text{Span}\{h_\lambda \mid \lambda \vdash k\}$ for any $\mu \vdash k$.

If $\mu = (k)$, using $\sum_{j \geq 0} (-1)^j e_j h_{k-j} = 0$, we have

$$e_k = \sum_{j=0}^{k-1} (-1)^{k+j-1} e_j h_{k-j}$$

Each e_j for $0 \leq j \leq k-1$ belongs to $\text{Span}\{h_\lambda \mid \lambda \vdash j\}$ by induction. Hence $e_k \in \text{Span}\{h_\lambda \mid \lambda \vdash k\}$.

If $\mu \neq (k)$, then $e_\mu = e_{\mu_1} e_{\mu_2} \dots e_{\mu_{\ell(\mu)}}$. Since each $\mu_i < k$, each factor $e_{\mu_i} \in \text{Span}\{h_\lambda \mid \lambda \vdash \mu_i\}$ by induction. The result follows. $\square$

2.12 The power sum symmetric functions

We can think of $e_k = m_{1^k}$, where (1^k) is the smallest partition in lexicographic order.

Question: What about the largest partition in lexicographic order, (k) ?

Definition 2.12.1. Let $k \geq 1$. The *k -th power sum symmetric function* p_k is defined as

$$p_k = m_{(k)} = \sum_{i \geq 1} x_i^k = x_1^k + x_2^k + \dots$$

And for any partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$, we define

$$p_\lambda = p_{\lambda_1} p_{\lambda_2} \dots p_{\lambda_\ell}$$

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Examples:

$$p_3 = x_1^3 + x_2^3 + x_3^3 + \cdots = m_3$$

$$\begin{aligned} p_{2,1} &= (x_1^2 + x_2^2 + x_3^2 + \cdots)(x_1 + x_2 + x_3 + \cdots) = x_1^3 + x_2^3 + \cdots + x_1^2 x_2 + x_1 x_2^2 + \cdots \\ &= m_{2,1} + m_3 \end{aligned}$$

$$\begin{aligned} p_{1,1,1} &= (x_1 + x_2 + \cdots)^3 = 6x_1 x_2 x_3 + \cdots + 3x_1^2 x_2 + \cdots + x_1^3 + \cdots \\ &= 6m_{1,1,1} + 3m_{2,1} + m_3 \end{aligned}$$

The p_λ are symmetric, so there exist coefficients $M_{\lambda,\mu}(p, m)$ such that

$$p_\lambda = \sum_{\mu \vdash |\lambda|} M_{\lambda,\mu}(p, m) m_\mu$$

2.13 The coefficients $M_{\lambda,\mu}(p, m)$

Fact: $p_k = m_k$ and $p_\lambda = p_{\lambda_1} p_{\lambda_2} \cdots p_{\lambda_{\ell(\lambda)}}$. Hence p_λ is symmetric.

For $k = 3$, the system

$$\begin{aligned} p_{1,1,1} &= 6m_{1,1,1} + 3m_{2,1} + m_3 \\ p_{2,1} &= m_{2,1} + m_3 \\ p_3 &= m_3 \end{aligned}$$

corresponds to the matrix $[M_{\lambda,\mu}(p, m)]$:

$$\begin{pmatrix} (1^3) & (2, 1) & (3) \end{pmatrix} \begin{pmatrix} (1^3) \\ (2, 1) \\ (3) \end{pmatrix} \begin{bmatrix} 6 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Proposition 2.13.1. $M_{\lambda,\mu}(p, m)$ is equal to the number of tableaux of shape λ with *constant* rows of positive integers that have content μ .

Example: Find the coefficient of $m_{5,4,3}$ in $p_{4,3,2,2,1}$.

The shape of the tableaux is $\lambda = (4, 3, 2, 2, 1)$ and the content is $\mu = (5, 4, 3)$ (five 1s, four 2s, three 3s). The valid tableaux with constant rows are:

1	1	1	1
3	3	3	
2	2		
2	2		
1			

2	2	2	2
1	1	1	
1	1		
3	3		
3			

2	2	2	2
1	1	1	
3	3		
1	1		
3			

2	2	2	2
3	3	3	
1	1		
1	1		
1			

Hence, $M_{(4,3,2,2,1),(5,4,3)}(p, m) = 4$.

Note: The matrix for $k = 3$ is upper triangular and has non-zero determinant.

2.14 The matrix $[M_{\lambda,\mu}(p, m)]$ is triangular

Proposition 2.14.1. *Let $\lambda, \mu \vdash k$ with $|\lambda| = |\mu|$. Then:*

- (i) *If $\lambda >_{\text{lex}} \mu$, then $M_{\lambda,\mu}(p, m) = 0$.*
- (ii) *$M_{\lambda,\lambda}(p, m) > 0$.*

Proof. (i) and (ii) Since $p_\lambda = p_{\lambda_1} p_{\lambda_2} \dots p_{\lambda_{\ell(\lambda)}}$ and $M_{\lambda,\mu}(p, m)$ is the coefficient of $x_1^{\mu_1} x_2^{\mu_2} \dots x_{\ell(\mu)}^{\mu_{\ell(\mu)}}$, we expand:

$$p_\lambda = (x_1^{\lambda_1} + x_2^{\lambda_1} + \dots)(x_1^{\lambda_2} + x_2^{\lambda_2} + \dots) \dots (x_1^{\lambda_{\ell(\lambda)}} + x_2^{\lambda_{\ell(\lambda)}} + \dots)$$

A factor in this product is obtained by picking one term $x_{i_j}^{\lambda_j}$ from the j -th factor. Suppose we pick $x_{i_1}^{\lambda_1}, x_{i_2}^{\lambda_2}, \dots, x_{i_{\ell(\lambda)}}^{\lambda_{\ell(\lambda)}}$. If $i_1, i_2, \dots, i_{\ell(\lambda)}$ are all distinct, then the product of these variables is $x_1^{\mu_1} x_2^{\mu_2} \dots x_{\ell(\mu)}^{\mu_{\ell(\mu)}}$ only if $\lambda = \mu$. We can form this product in $m_1(\lambda)! m_2(\lambda)! \dots m_{\ell(\lambda)}(\lambda)!$ ways, where $m_i(\lambda)$ is the multiplicity of i in λ . This number is strictly positive, hence $M_{\lambda,\lambda}(p, m) > 0$.

If $i_1, i_2, \dots, i_{\ell(\lambda)}$ are not all distinct, then some μ_i 's are sums of λ_j 's. This implies that there is a first part in μ that is strictly greater than the corresponding part in λ , so $\mu >_{\text{lex}} \lambda$. Thus, if $\lambda >_{\text{lex}} \mu$, we have $M_{\lambda,\mu}(p, m) = 0$. $\square$

2.15 The power sum symmetric functions form a basis

Proposition 2.15.1. *The set $\{p_\lambda \mid \lambda \vdash k\}$ is a basis for Λ^k , called the **power sum symmetric basis**.*

Proof. Let A be the $p(k) \times p(k)$ matrix with rows and columns indexed by partitions of k ordered lexicographically, with entries $M_{\lambda,\mu}(p, m)$. Then A is upper triangular with non-zero diagonal entries. Hence A is invertible, so it is the change of basis matrix from the p basis to the m basis. $\square$

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Example: For $k = 4$, the change of basis matrix A (with rows and columns ordered $(1^4), (2, 1^2), (2^2), (3, 1), (4)$) is:

$$A = \begin{bmatrix} 24 & 12 & 6 & 4 & 1 \\ 0 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Generating Function for Power Sum Symmetric Functions:

$$P(t) = \sum_{k=1}^{\infty} \frac{p_k}{k} t^k = \log \left(\prod_{i=1}^{\infty} \frac{1}{1 - x_i t} \right)$$

Proof. From calculus, $\log \left(\frac{1}{1-x} \right) = \sum_{n=1}^{\infty} \frac{1}{n} x^n$. Therefore,

$$P(t) = \sum_{k=1}^{\infty} \frac{p_k}{k} t^k = \sum_{k=1}^{\infty} \sum_{i=1}^{\infty} \frac{1}{k} (x_i t)^k = \sum_{i=1}^{\infty} \log \left(\frac{1}{1 - x_i t} \right) = \log \left(\prod_{i=1}^{\infty} \frac{1}{1 - x_i t} \right)$$

□

Chapter 3

Schur Symmetric Functions

Lecture 3: Schur Symmetric Functions

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3.1 Definitions and Tableaux

Definition 3.1.1. Let λ be a partition. A **Young tableau** (or simply tableau) T of shape λ is a filling of the boxes of the Young diagram of λ with positive integers. The **weight** (or content) of T , denoted x^T , is defined by

$$x^T = \prod_{i \in T} x_i = x_1^{c_1} x_2^{c_2} \cdots$$

where c_i is the number of times i appears in T .

We define three special classes of tableaux of shape λ . In each case there is no condition relating entries in different rows:

- ◇ $\text{StrictRow}(\lambda)$: The set of tableaux that are **strictly increasing** along each row.
- ◇ $\text{WeakRow}(\lambda)$: The set of tableaux that are **weakly increasing** along each row.
- ◇ $\text{ConstantRow}(\lambda)$: The set of tableaux that are **constant** along each row.

3. SCHUR SYMMETRIC FUNCTIONS

Proposition 3.1.2. *The elementary, homogeneous, and power sum symmetric functions can be expressed as sums over tableaux:*

$$e_\lambda = \sum_{T \in \text{StrictRow}(\lambda)} x^T, \quad h_\lambda = \sum_{T \in \text{WeakRow}(\lambda)} x^T, \quad p_\lambda = \sum_{T \in \text{ConstantRow}(\lambda)} x^T$$

3.2 Schur Functions Definition

Definition 3.2.1. A **semi-standard Young tableau (SSYT)** of shape λ is a filling of the boxes of λ with positive integers such that:

- (a) Entries are **weakly increasing** across each row from left to right.
- (b) Entries are **strictly increasing** down each column from top to bottom.

We write $SST(\lambda)$ for the set of all semi-standard Young tableaux of shape λ , and $SST(\lambda, \mu)$ for the set of semi-standard Young tableaux of shape λ with content μ .

Definition 3.2.2. For any partition λ , the **Schur function** s_λ is defined by

$$s_\lambda = \sum_{T \in SST(\lambda)} x^T$$

Example 3.2.3. Compute $s_{2,1}(x_1, x_2, x_3)$. The shape $\lambda = (2, 1)$ has 3 boxes. The semi-standard Young tableaux filled with $\{1, 2, 3\}$ are:

$$\begin{array}{|c|c|} \hline 1 & 1 \\ \hline 2 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 3 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 2 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 3 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 2 & 2 \\ \hline 3 & \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 2 & 3 \\ \hline 3 & \\ \hline \end{array}$$

Evaluating their weights yields:

$$s_{2,1}(x_1, x_2, x_3) = m_{2,1}(x_1, x_2, x_3) + 2m_{1,1,1}(x_1, x_2, x_3)$$

3.3 Special Cases: s_{1^k} and s_k

Proposition 3.3.1. *For any positive integer k :*

(a) $s_{1^k} = e_k$.

(b) $s_k = h_k$.

Proof. (a) The shape (1^k) is a single column of k boxes. A filling is semi-standard if and only if the entries strictly increase from top to bottom: $1 \leq i_1 < i_2 < \cdots < i_k$. Thus,

$$s_{1^k} = \sum_{1 \leq i_1 < i_2 < \cdots < i_k} x_{i_1} x_{i_2} \cdots x_{i_k} = e_k$$

(b) The shape (k) is a single row of k boxes. A filling is semi-standard if and only if the entries weakly increase from left to right: $1 \leq i_1 \leq i_2 \leq \cdots \leq i_k$. Thus,

$$s_k = \sum_{1 \leq i_1 \leq i_2 \leq \cdots \leq i_k} x_{i_1} x_{i_2} \cdots x_{i_k} = h_k$$

□

3.4 Symmetry of Schur Functions

Theorem 3.4.1. *For any partition λ , the Schur function s_λ is a symmetric function, i.e., $s_\lambda \in \Lambda$.*

Outline of Proof via Bender-Knuth Involution. To show that s_λ is symmetric, it suffices to show that swapping adjacent variables x_i and x_{i+1} leaves s_λ invariant. This is done by defining an involution ϕ_i on $SST(\lambda)$ that maps a tableau of weight $(\dots, \mu_i, \mu_{i+1}, \dots)$ to a tableau of weight $(\dots, \mu_{i+1}, \mu_i, \dots)$.

In any tableau $T \in SST(\lambda)$, call an entry i or $i+1$ *matched* if they appear in the same column (so i is directly above $i+1$). All other entries i and $i+1$ are *free*. In each row, the free entries form a contiguous block of i 's followed by $(i+1)$'s. The involution ϕ_i replaces a block of a i 's and b $(i+1)$'s with b i 's and a $(i+1)$'s. This preserves semi-standardness and bijectively pairs tableaux with weight x^T and $x^{\phi_i(T)}$, proving $|SST(\lambda, \mu)| = |SST(\lambda, \tilde{\mu})|$. □

3.5 The Kostka Numbers

Definition 3.5.1. The **Kostka numbers** $K_{\lambda,\mu}$ are defined by

$$s_\lambda = \sum_{\mu \vdash |\lambda|} K_{\lambda,\mu} m_\mu$$

Equivalently, $K_{\lambda,\mu} = |SST(\lambda, \mu)|$ is the number of semi-standard Young tableaux of shape λ and content μ .

Example 3.5.2. For $k = 3$, ordering partitions lexicographically as $(3), (2, 1), (1, 1, 1)$, the Kostka matrix $K^{(3)} = (K_{\lambda,\mu})$ is:

$$K^{(3)} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

For $k = 4$, ordering partitions as $(4), (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1)$, the Kostka matrix $K^{(4)}$ is:

$$K^{(4)} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

3.6 Schur Functions Form a Basis

Theorem 3.6.1. For any partitions $\lambda, \mu \vdash k$:

(a) $K_{\lambda,\mu} = 0$ if $\mu >_{\text{lex}} \lambda$.

(b) $K_{\lambda,\lambda} = 1$.

Consequently, the set $\{s_\lambda \mid \lambda \vdash k\}$ forms a basis for the space of homogeneous symmetric functions of degree k , Λ^k .

Proof. If $\mu >_{\text{lex}} \lambda$, there is no semi-standard tableau of shape λ and content μ , because the entries $1, 2, \dots, i$ must all fit into the first i rows, which is impossible if $\mu_1 + \dots + \mu_i > \lambda_1 + \dots + \lambda_i$. For $\mu = \lambda$, the unique SSYT of shape λ and content λ has row i filled entirely with i , so $K_{\lambda,\lambda} = 1$. Thus the transition matrix $(K_{\lambda,\mu})$ is upper triangular with 1s on the diagonal, so it is invertible. $\square$

3.7 The Jacobi-Trudi Identity

Theorem 3.7.1 (Jacobi-Trudi Identity). *Let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$ be a partition. Then the Schur function s_λ can be written as a determinant of complete homogeneous symmetric functions:*

$$s_\lambda = \det [h_{\lambda_i + j - i}]_{1 \leq i, j \leq k}$$

where by convention $h_0 = 1$ and $h_m = 0$ for $m < 0$.

In matrix form:

$$s_\lambda = \begin{vmatrix} h_{\lambda_1} & h_{\lambda_1+1} & h_{\lambda_1+2} & \cdots & h_{\lambda_1+k-1} \\ h_{\lambda_2-1} & h_{\lambda_2} & h_{\lambda_2+1} & \cdots & h_{\lambda_2+k-2} \\ h_{\lambda_3-2} & h_{\lambda_3-1} & h_{\lambda_3} & \cdots & h_{\lambda_3+k-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h_{\lambda_k-k+1} & h_{\lambda_k-k+2} & h_{\lambda_k-k+3} & \cdots & h_{\lambda_k} \end{vmatrix}$$

3.8 Jacobi-Trudi Determinant Examples

Example 3.8.1. (a) For $\lambda = (2, 1)$ with $k = 2$:

$$s_{2,1} = \begin{vmatrix} h_2 & h_3 \\ h_0 & h_1 \end{vmatrix} = h_2 h_1 - h_3 h_0 = h_{2,1} - h_3$$

(b) For $\lambda = (3, 2, 1)$ with $k = 3$:

$$s_{3,2,1} = \begin{vmatrix} h_3 & h_4 & h_5 \\ h_1 & h_2 & h_3 \\ h_{-1} & h_0 & h_1 \end{vmatrix} = \begin{vmatrix} h_3 & h_4 & h_5 \\ h_1 & h_2 & h_3 \\ 0 & 1 & h_1 \end{vmatrix}$$

3.9 The Omega Involution

Definition 3.9.1. The **omega map** $\omega : \Lambda \rightarrow \Lambda$ is the algebra homomorphism defined on the elementary generators by

$$\omega(e_n) = h_n \quad \text{for all } n \geq 1$$

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Proposition 3.9.2 (Properties of ω). (a) ω is an involution: $\omega^2 = \text{id}$.

(b) $\omega(e_\lambda) = h_\lambda$ and $\omega(h_\lambda) = e_\lambda$.

(c) $\omega(s_\lambda) = s_{\lambda'}$, where λ' is the conjugate partition of λ .

(d) $\omega(p_\lambda) = (-1)^{|\lambda| - \ell(\lambda)} p_\lambda$.

Applying ω to the Jacobi-Trudi identity yields the **Dual Jacobi-Trudi Identity**:

$$s_{\lambda'} = \det [e_{\lambda_i + j - i}]_{1 \leq i, j \leq k} \iff s_\lambda = \det [e_{\lambda'_i + j - i}]_{1 \leq i, j \leq \ell(\lambda')}$$

3.10 Writing s_λ in the Elementary Basis

Corollary 3.10.1. The Schur function s_λ can be expanded in terms of the elementary symmetric functions using the dual Jacobi-Trudi formula:

$$s_\lambda = \det [e_{\lambda'_i + j - i}]_{1 \leq i, j \leq \ell(\lambda')}$$

Example 3.10.2. Compute $s_{3,2}$ in the elementary basis. The conjugate of $\lambda = (3, 2)$ is $\lambda' = (2, 2, 1)$, which has length 3. Thus:

$$s_{3,2} = \begin{vmatrix} e_2 & e_3 & e_4 \\ e_1 & e_2 & e_3 \\ e_{-1} & e_0 & e_1 \end{vmatrix} = \begin{vmatrix} e_2 & e_3 & e_4 \\ e_1 & e_2 & e_3 \\ 0 & 1 & e_1 \end{vmatrix}$$

Expanding along the last row yields:

$$s_{3,2} = -1 \cdot (e_2 e_3 - e_1 e_4) + e_1 (e_2^2 - e_1 e_3) = e_1 e_2^2 - e_1^2 e_3 - e_2 e_3 + e_1 e_4$$

3.11 The Forgotten Basis

Definition 3.11.1. The forgotten symmetric functions f_λ are defined by applying the ω involution to the monomial basis:

$$f_\lambda = \omega(m_\lambda)$$

Proposition 3.11.2. *The set $\{f_\lambda \mid \lambda \vdash k\}$ forms a basis for Λ^k .*

Example 3.11.3. $\diamond f_{2,1} = \omega(m_{2,1}) = -2m_3 - m_{2,1}$

$$\diamond f_{1,1,1} = \omega(m_{1,1,1}) = m_{1,1,1} + m_{2,1} + m_3$$

$$\diamond f_3 = \omega(m_3) = m_3$$

$$\diamond f_{3,2} = \omega(m_{3,2}) = -m_{3,2} - 2m_5$$

3.12 The Pieri Rules: Definitions

Definition 3.12.1. Let $\mu \subseteq \lambda$ be partitions. The **skew diagram** λ/μ is the set of boxes belonging to λ but not to μ .

Definition 3.12.2. A **horizontal strip** is a skew diagram λ/μ that contains **at most one box in each column**.

Note: Skew diagrams do not need to form a single connected piece.

3.13 The Pieri Rules for h_k

Theorem 3.13.1 (Pieri Rule for h_k). *For any partition μ and non-negative integer k :*

$$h_k s_\mu = \sum_{\lambda} s_\lambda$$

where the sum is over all partitions $\lambda \supseteq \mu$ such that λ/μ is a horizontal strip of size k .

Example 3.13.2. Let $\mu = (2, 2)$:

$$\diamond h_1 s_{2,2} = s_{3,2} + s_{2,2,1}$$

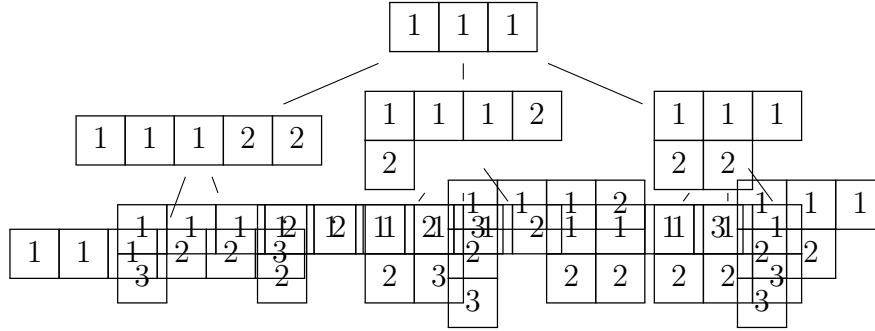
$$\diamond h_2 s_{2,2} = s_{4,2} + s_{3,2,1} + s_{2,2,2}$$

$$\diamond h_3 s_{2,2} = s_{5,2} + s_{4,2,1} + s_{3,2,2}$$

3.14 Expanding h_μ in the Schur Basis

To write $h_\mu = h_{\mu_1} h_{\mu_2} \dots h_{\mu_\ell}$ in the Schur basis, we repeatedly apply the Pieri rule for horizontal strips, generating a tree of tableaux.

Example 3.14.1. Expand $h_{3,2,1}$ in the Schur basis:



Summing the shapes of the leaves gives:

$$h_{3,2,1} = s_6 + 2s_{5,1} + 2s_{4,2} + s_{4,1,1} + s_{3,3} + s_{3,2,1}$$

3.15 The Pieri Rules for e_k

Definition 3.15.1. A **vertical strip** is a skew diagram λ/μ that contains **at most one box in each row**.

Theorem 3.15.2 (Pieri Rule for e_k). For any partition μ and non-negative integer k :

$$e_k s_\mu = \sum_{\lambda} s_\lambda$$

where the sum is over all partitions $\lambda \supseteq \mu$ such that λ/μ is a vertical strip of size k .

Example 3.15.3. Let $\mu = (2, 2)$:

$$\diamond e_2 s_{2,2} = s_{3,3} + s_{3,2,1} + s_{2,2,1,1}$$

$$\diamond e_3 s_{2,2} = s_{3,3,1} + s_{3,2,1,1} + s_{2,2,1,1,1}$$

3.16 Expanding e_μ in the Schur Basis

To write $e_\mu = e_{\mu_1} e_{\mu_2} \dots e_{\mu_\ell}$ in the Schur basis, we repeatedly apply the Pieri rule for vertical strips.

Example 3.16.1. Expand $e_{3,2,1}$ in the Schur basis:

$$e_{3,2,1} = s_{3,2,1} + s_{2^3} + 2s_{2^2,1^2} + s_{3,1^3} + 2s_{2,1^4} + s_{1^6}$$

3.17 The Murnaghan-Nakayama Rule

Definition 3.17.1. A **border strip** (or ribbon) is a connected skew diagram λ/μ that contains no 2×2 square of boxes.

The **height** $h(B)$ of a border strip $B = \lambda/\mu$ is defined as

$$h(B) = (\text{number of rows of } B) - 1$$

Theorem 3.17.2 (Murnaghan-Nakayama Rule). *For any integer $k \geq 1$ and partition μ :*

$$p_k s_\mu = \sum_{\lambda} (-1)^{h(\lambda/\mu)} s_\lambda$$

where the sum is over all partitions $\lambda \supseteq \mu$ such that λ/μ is a border strip of size k .

Example 3.17.3. Compute $p_3 s_{6,3,3,2,1}$:

$$p_3 s_{6,3,3,2,1} = s_{9,3,3,2,1} + s_{6,6,3,2,1} - s_{6,5,4,2,1} - s_{6,3,3,2,2,2} - s_{6,3,3,3,3} + s_{6,3,3,2,1,1,1,1}$$

Chapter 4

The Chromatic Symmetric Function

Lecture 4: The Chromatic Symmetric Function

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CIMPA-ECCO Summer School, August 6, 2026

4.1 Introduction and Graphs

Definition 4.1.1. A **graph** $G = (V, E)$ consists of a finite set of **vertices** (or nodes) $V = V(G)$ and a set of **edges** $E = E(G)$, where each edge is an unordered pair $\{u, v\}$ of distinct vertices. Vertices represent objects, and edges represent relationships between pairs of objects.

Definition 4.1.2. A **cycle** C_k is a closed walk of k distinct vertices $v_1, v_2, \dots, v_k$ such that $\{v_i, v_{i+1}\} \in E$ for all $1 \leq i \leq k-1$ and $\{v_k, v_1\} \in E$.

A graph G is **connected** if there is a path between every pair of vertices. A **tree** is a connected graph with no cycles. Vertices of degree 1 in a tree are called **leaves**.

4.2 Colorings and the Chromatic Polynomial

Definition 4.2.1. A **proper coloring** of a simple graph $G = (V, E)$ with n colors is a map $\kappa : V \rightarrow \{1, 2, \dots, n\}$ such that if $\{u, v\} \in E$, then $\kappa(u) \neq \kappa(v)$.

Definition 4.2.2. The **chromatic polynomial** $\chi_G(n)$ counts the number of proper colorings of G using at most n colors.

Example 4.2.3. (a) For the triangle graph C_3 :

$$\chi_{C_3}(n) = n(n-1)(n-2)$$

For $n = 3$ colors, there are $\chi_{C_3}(3) = 3 \cdot 2 \cdot 1 = 6$ proper colorings.

(b) For any tree T with k vertices:

$$\chi_T(n) = n(n-1)^{k-1}$$

Note: The chromatic polynomial $\chi_G(n)$ cannot distinguish between non-isomorphic trees on the same number of vertices! For example, all trees on k vertices have the exact same chromatic polynomial $\chi_T(n) = n(n-1)^{k-1}$.

4.3 The Chromatic Symmetric Function

To capture more structural information about a graph, Richard Stanley (1995) introduced a symmetric function generalization of the chromatic polynomial.

Definition 4.3.1 (Stanley, 1995). Let $G = (V, E)$ be a simple graph with vertex set $V = \{v_1, v_2, \dots, v_n\}$. To each proper coloring $\kappa : V \rightarrow \mathbb{N}$, associate the monomial $x^\kappa = x_{\kappa(v_1)}x_{\kappa(v_2)} \cdots x_{\kappa(v_n)}$.

The **chromatic symmetric function (CSF)** of G is defined as

$$\mathbf{X}_G(x_1, x_2, \dots) = \sum_{\kappa \text{ proper}} x_{\kappa(v_1)}x_{\kappa(v_2)} \cdots x_{\kappa(v_n)}$$

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Proposition 4.3.2 (Properties of the CSF). (a) *Evaluation to Chromatic Polynomial:*

$$\mathbf{X}_G(\underbrace{1, 1, \dots, 1}_n, 0, 0, \dots) = \chi_G(n)$$

(b) *Disjoint Union:* If $G = H_1 \sqcup H_2$ is the disjoint union of graphs H_1 and H_2 , then

$$\mathbf{X}_G = \mathbf{X}_{H_1} \mathbf{X}_{H_2}$$

(c) *Symmetry:* $\mathbf{X}_G \in \Lambda$ is a homogeneous symmetric function of degree $|V(G)|$.

4.4 The CSF in the Power Basis

For a subset of edges $S \subseteq E(G)$, let $\pi(S)$ be the integer partition whose parts are the vertex-sizes of the connected components of the spanning subgraph $(V(G), S)$.

Theorem 4.4.1 (Stanley, 1995). *For any graph $G = (V, E)$:*

$$\mathbf{X}_G = \sum_{S \subseteq E(G)} (-1)^{|S|} p_{\pi(S)}$$

where p_λ is the power sum symmetric function.

Example 4.4.2. Let G be the 5-vertex graph formed by a triangle with two attached leaves (a "bull-like" graph). Summing over all $2^{|E|}$ edge subsets $S \subseteq E(G)$ according to component sizes yields:

$$\mathbf{X}_G = p_1^5 - 5p_{2,1^3} + 2p_{3,1^2} + 7p_{2^2,1} - 4p_{3,2} - 2p_{4,1} + p_5$$

4.5 Graph Invariants Recoverable from the CSF

Expanding the CSF in the power basis $\mathbf{X}_G = \sum_{\lambda \vdash |V(G)|} a_\lambda p_\lambda$, many structural properties of G can be extracted directly from the coefficients a_λ :

Proposition 4.5.1 (Recoverable Invariants). *Let $G = (V, E)$ with $|V| = n$.*

◇ *Number of Vertices:* $|V| = |\lambda|$.

◇ *Number of Edges:* $|E| = -a_{(2,1^{n-2})}$.

◇ *Number of Matchings of Size k : Given by $(-1)^k a_{(2^k, 1^{n-2k})}$.*

◇ *Number of Triangles: [Orellana–Scott] The number of triangles in G is*

$$\# \text{ Triangles} = \binom{|E|}{2} - a_{(2,2,1^{n-4})} - a_{(3,1^{n-3})}$$

◇ *Tree Recovery: [Martin–Morin–Wagner] For trees, many parameters including the degree sequence can be recovered.*

4.6 The Star Basis of Symmetric Functions

Definition 4.6.1. Let St_k denote the star graph on k vertices (one central vertex of degree $k - 1$ connected to $k - 1$ leaves). The chromatic symmetric function of St_k is

$$\mathbf{st}_k := \mathbf{X}_{St_k} = \sum_{i=0}^{k-1} (-1)^i \binom{k-1}{i} p_{(i+1, 1^{k-1-i})}$$

For any partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$, define

$$\mathbf{st}_\lambda := \mathbf{st}_{\lambda_1} \mathbf{st}_{\lambda_2} \dots \mathbf{st}_{\lambda_\ell}$$

Example 4.6.2. For St_4 (star graph on 4 vertices):

$$\mathbf{X}_{St_4} = -p_4 + 3p_{3,1} - 3p_{2,1,1} + p_{1,1,1,1} = \mathbf{st}_4$$

Theorem 4.6.3 (Cho–van Willigenburg, 2016). *The set $\{\mathbf{st}_\lambda \mid \lambda \vdash n\}$ forms a basis for the space of homogeneous symmetric functions of degree n , Λ^n .*

Example 4.6.4 (Trees in Star Basis). ◇ Path P_4 : $\mathbf{X}_{P_4} = \mathbf{st}_4 - \mathbf{st}_{3,1} + \mathbf{st}_{2,2}$

◇ Star St_4 : $\mathbf{X}_{St_4} = \mathbf{st}_4$

Example 4.6.5 (Unicyclic Graphs in Star Basis). ◇ Triangle C_3 : $\mathbf{X}_{C_3} = 2\mathbf{st}_3 - \mathbf{st}_{2,1}$

◇ Dart graph: $\mathbf{X} = 2\mathbf{st}_5 - 2\mathbf{st}_{4,1} + \mathbf{st}_{4,2}$

From the star basis expansion, the size of the cycle in a unicyclic graph can be recovered [Bingham, Johnston, Lawson, Orellana, Pan, and Sato (2025)].

4.7 Graph Isomorphism and the Tree Isomorphism Conjecture

Definition 4.7.1. Two graphs G and H are **isomorphic** ($G \cong H$) if there exists a bijection $\phi : V(G) \rightarrow V(H)$ such that $\{u, v\} \in E(G)$ if and only if $\{\phi(u), \phi(v)\} \in E(H)$.

Isomorphic graphs always have the same chromatic symmetric function ($X_G = X_H$). However, Stanley (1995) showed that non-isomorphic graphs can share the same CSF.

Theorem 4.7.2 (Open Problem: Tree Isomorphism Conjecture). *If T_1 and T_2 are non-isomorphic trees, then $X_{T_1} \neq X_{T_2}$. In other words, the chromatic symmetric function distinguishes trees up to isomorphism.*

Known Progress:

- ◇ Verified for all trees with $n \leq 29$ vertices (Heil and Ji, 2019 — over 5.46 billion trees).
- ◇ Verified for caterpillar graphs (Loebl, Sereni, 2018).
- ◇ Verified for "proper" trees with diameter ≤ 5 (Aliste-Prieto, De Mier, Orellana, Zamora, 2022).

4.8 Proper Trees, Core, and Edge Operations

Definition 4.8.1. Let T be a tree.

- ◇ A **leaf** is a vertex of degree 1. A **leaf edge** is an edge incident to a leaf.
- ◇ An **internal vertex** is a vertex of degree > 1 .
- ◇ An **internal edge** is an edge whose endpoints are both internal vertices.
- ◇ A **proper tree** is a tree where every vertex is a leaf or adjacent to at least one leaf.
- ◇ The **core** of a tree is the weighted graph obtained by removing all leaves and labeling each remaining (internal) vertex with the number of leaves adjacent to it plus one.

Edge Operations on Graphs:

- ◇ **Deletion** ($G - e$): Delete edge e from G .
- ◇ **Dot-contraction** $((G/e) + v)$: Contract edge e and add an isolated vertex v .
- ◇ **Leaf-contraction** $((G/e) + v + e')$: Contract edge e , add isolated vertex v , and add a leaf edge e' to the contracted vertex.

4.9 Deletion-Near-Contraction (DNC)

Theorem 4.9.1 (Aliste-Prieto, De Mier, Orellana, Zamora). *For any graph G and any edge e :*

$$\mathbf{X}_G = \mathbf{X}_{G-e} - \mathbf{X}_{(G/e)+v} + \mathbf{X}_{(G/e)+v+e'}$$

Notes on DNC. ◇ DNC is applied only to **internal edges**.

- ◇ The only simple, connected graphs without internal edges are **star graphs**.
- ◇ Applying DNC recursively to internal edges computes the expansion of $\mathbf{X}_G$ in the **star basis** $\mathbf{st}_\lambda$.
- ◇ **No Cancellations:** Each branch in the DNC recursion tree contributes directly to a non-zero term in the star expansion!

□

Example 4.9.2 (DNC Expansion of a Tree). Applying DNC to a tree with internal edges yields:

$$\mathbf{X}_T = \mathbf{st}_{4,2,2} - \mathbf{st}_{4,3,1} + \mathbf{st}_{4,4} - \mathbf{st}_{5,2,1} + \mathbf{st}_{6,1,1} + \mathbf{st}_{6,2} - 2\mathbf{st}_{7,1} + \mathbf{st}_8$$

4.10 Properties and Leading Partitions in the Star Basis

Expanding $\mathbf{X}_G = \sum_{\lambda \vdash |V|} c_\lambda \mathbf{st}_\lambda$ in the star basis:

Proposition 4.10.1. (a) G is connected if and only if $c_{(n)} \neq 0$ (i.e. $\mathbf{st}_{(n)}$ appears with non-zero coefficient).

(b) If $G = T$ is a tree, then $c_{(n)} = 1$.

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(c) For hook partitions $(n - m, 1^m)$, the coefficient is:

$$c_{n-m, 1^m} = (-1)^m \binom{I(T)}{m}$$

where $I(T)$ is the number of internal edges in T . In particular, $|c_{n-m, 1}| = I(T)$.

Definition 4.10.2. The **leaf component partition** $\lambda_{LC(T)}$ of a tree T is the integer partition whose parts are the vertex-orders of the connected components of $T \setminus I(T)$ (deleting all internal edges).

The **leading partition** λ_{lead} of $\mathbf{X}_T$ is the smallest partition in lexicographic order occurring with non-zero coefficient in the star-expansion.

Theorem 4.10.3 (Gonzalez-Orellana-Tomba). For any tree T , the leading partition equals the leaf component partition:

$$\lambda_{lead} = \lambda_{LC}$$

Furthermore, the leading coefficient $c_{\lambda_{lead}}$ is given by

$$c_{\lambda_{lead}} = (-1)^k \prod_{u_i} (\deg(u_i) - 1)$$

where the product is over all vertices u_i of degree greater than 1 in the core of T having label 1, and k is the number of such vertices.

Corollary 4.10.4. For all proper trees, $c_{\lambda_{lead}} = 1$ (since proper trees have no core vertices with label 1).

Part II

Worksheets and Solutions

Chapter 5

Worksheets

5.1 Worksheet 1

1. Find all permutations $\pi \in S_4$ for which $\pi(f) = f$, where $f(x_1, x_2, x_3, x_4) = x_1x_2^2x_4^4 + x_2^2x_3^4x_4 + x_1^4x_2^2x_3$.
2. Find a polynomial f in x_1, x_2, x_3 which has $\pi(f) = \text{sgn}(\pi)f$ for all $\pi \in S_3$, and which has $x_1^7x_3^2$ as one of its terms. Recall that $\text{sgn}(\pi)$ is the **sign** of the permutation that is 1, if π is the product of an even number of transpositions and -1 if π is the product of an odd number of permutations.
3. Compute the four monomial symmetric polynomials $m_{3,2}(x_1, x_2)$, $m_{3,2}(x_1, x_2, x_3)$, $m_{3,3,1,1}(x_1, x_2, x_3)$, and $m_{3,3,1,1}(x_1, x_2, x_3, x_4)$.
4. Compute the product of the formal power series $f(x) = \sum_{i=0}^{\infty} x^i$ and $g(x) = \sum_{j=0}^{\infty} jx^j$.
5. Compute the product of the formal power series $f = \sum_{i=1}^{\infty} x_ix_{i+1}$ and $g = \sum_{j=1}^{\infty} x_j$.
6. Compute $m_{2,1}$. Is $m_2m_1 = m_{2,1}$?
7. Using symmetry, write $m_3m_2m_1$ as a linear combination of $\{m_\lambda : \lambda \vdash 6\}$.
8. Without multiplying everything out, find the coefficient of $m_{2,1^{a+b}}$ in the product $m_{1^{a+1}}m_{1^{b+1}}$.
9. Write the following as a sum of monomial symmetric functions:
 - (a) $\prod_{i=1}^{\infty} (1 + x_i)$.
 - (b) $\prod_{i=1}^{\infty} (1 - x_i)$

5. WORKSHEETS

10. Use the geometric series to write a very simple infinite product using the variables x_i 's that is equal to the sum

$$\sum_{\lambda} m_{\lambda},$$

where here the sum is over all partitions of all positive integers.

Español by Ignacio Rojas

- Encuentre todas las permutaciones $\pi \in S_4$ para las cuales $\pi(f) = f$, donde $f(x_1, x_2, x_3, x_4) = x_1x_2^2x_4^4 + x_2^2x_3^4x_4 + x_1^4x_2^2x_3$.
- Encuentre un polinomio f en x_1, x_2, x_3 que satisfaga $\pi(f) = \text{sgn}(\pi)f$ para toda $\pi \in S_3$, y que tenga $x_1^7x_3^2$ como uno de sus términos. Recuerde que $\text{sgn}(\pi)$ es el **signo** de la permutación, que es 1 si π es el producto de un número par de transposiciones y -1 si π es el producto de un número impar de permutaciones.
- Calcule los cuatro polinomios simétricos monomiales $m_{3,2}(x_1, x_2)$, $m_{3,2}(x_1, x_2, x_3)$, $m_{3,3,1,1}(x_1, x_2, x_3)$ y $m_{3,3,1,1}(x_1, x_2, x_3, x_4)$.
- Calcule el producto de las series de potencias formales $f(x) = \sum_{i=0}^{\infty} x^i$ y $g(x) = \sum_{j=0}^{\infty} jx^j$.
- Calcule el producto de las series de potencias formales $f = \sum_{i=1}^{\infty} x_i x_{i+1}$ y $g = \sum_{j=1}^{\infty} x_j$.
- Calcule $m_{2,1}$. ¿Vale que $m_2 m_1 = m_{2,1}$?
- Usando la simetría, escriba $m_3 m_2 m_1$ como una combinación lineal de $\{m_{\lambda} : \lambda \vdash 6\}$.
- Sin multiplicar todo, encuentre el coeficiente de $m_{2,1^{a+b}}$ en el producto $m_{1^{a+1}} m_{1^{b+1}}$.
- Reescriba lo siguiente como una suma de funciones simétricas monomiales:
 - $\prod_{i=1}^{\infty} (1 + x_i)$.
 - $\prod_{i=1}^{\infty} (1 - x_i)$
- Utilizando la serie geométrica, escriba un producto infinito muy sencillo usando las variables x_i que sea igual a la suma

$$\sum_{\lambda} m_{\lambda},$$

donde aquí la suma es sobre todas las particiones de todos los enteros positivos.

5.2 Worksheet 2

1. Compute $M_{\lambda,\mu}(e, m)$ for $\lambda = (2^2, 1^4)$ and $\mu = (3^2, 1^2)$.
2. Find and prove a formula for $M_{\lambda,1^n}(e, m)$, for any $\lambda \vdash n$.
3. Find and prove a formula for $M_{\lambda,n}(e, m)$, for any $\lambda \vdash n$.
4. Find and prove a formula for $M_{\lambda,(n-1,1)}(e, m)$, for any $\lambda \vdash n$.
5. Prove or disprove: If λ and μ are partitions with $|\lambda| = |\mu|$ and $\lambda >_{\text{lex}} \mu$, then $\mu' >_{\text{lex}} \lambda'$.
6. Write the formal power series

$$\prod_{i=1}^{\infty} (1 + x_i + x_i^2)$$

in terms of the elementary symmetric functions.

7. There is another combinatorial interpretation for $M_{\lambda,\mu}(e, m)$, if $\lambda, \mu \vdash k$, $M_{\lambda,\mu}(e, m)$ is equal to the number of $k \times k$ matrices with entries 0 or 1 only, such that the sum of the i -th row is μ_i and the sum of column j is λ_j for all i and j .
 - (a) Use this combinatorial interpretation to find $M_{\lambda,\mu}(e, m)$ for $\lambda = (4, 2, 1)$ and $\mu = (3, 1^4)$.
 - (b) Use this interpretation to show that $M_{\lambda,\mu}(e, m) = M_{\mu,\lambda}(e, m)$.
 - (c) Describe a bijection from the tableaux interpretation to the $\{0, 1\}$ -matrix interpretation.
8. Compute $M_{\lambda,\mu}(h, m)$ for $\lambda = (2^2, 1^4)$ and $\mu = (3^2, 1^2)$.
9. Find and prove a formula for $M_{\lambda,1^n}(h, m)$, for any $\lambda \vdash n$.
10. Find and prove a formula for $M_{\lambda,n}(h, m)$, for any $\lambda \vdash n$.
11. Find and prove a formula for $M_{\lambda,(n-1,1)}(h, m)$, for any $\lambda \vdash n$.
12. Show that for all $k \geq 1$ and all partitions $\lambda, \mu \vdash k$, the quantity $M_{\lambda,\mu}(p, m)$ is the number of fillings of the Young diagram of λ with positive integers for which the entries in each row are constant, and each integer i appears μ_i times.
13. Compute $M_{\lambda,\mu}(p, m)$ for $\lambda = (2^2, 1^4)$ and $\mu = (3^2, 1^2)$.

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14. Find and prove a formula for $M_{\lambda,1^n}(p, m)$, for any $\lambda \vdash n$.
15. Find and prove a formula for $M_{\lambda,n}(p, m)$, for any $\lambda \vdash n$.
16. Find and prove a formula for $M_{\lambda,(n-1,1)}(p, m)$, for any $\lambda \vdash n$.

Español

1. Calcule $M_{\lambda,\mu}(e, m)$ para $\lambda = (2^2, 1^4)$ y $\mu = (3^2, 1^2)$.
2. Encuentre y pruebe una fórmula para $M_{\lambda,1^n}(e, m)$, para cualquier $\lambda \vdash n$.
3. Encuentre y pruebe una fórmula para $M_{\lambda,n}(e, m)$, para cualquier $\lambda \vdash n$.
4. Encuentre y pruebe una fórmula para $M_{\lambda,(n-1,1)}(e, m)$, para cualquier $\lambda \vdash n$.
5. Pruebe o dé un contraejemplo: Si λ y μ son particiones con $|\lambda| = |\mu|$ y $\lambda >_{\text{lex}} \mu$, entonces $\mu' >_{\text{lex}} \lambda'$.
6. Escriba la serie de potencias formal

$$\prod_{i=1}^{\infty} (1 + x_i + x_i^2)$$

en términos de las funciones simétricas elementales.

7. Existe otra interpretación combinatoria para $M_{\lambda,\mu}(e, m)$: si $\lambda, \mu \vdash k$, $M_{\lambda,\mu}(e, m)$ es igual al número de matrices $k \times k$ cuyas entradas son únicamente 0 o 1, tales que la suma de la i -ésima fila es μ_i y la suma de la columna j es λ_j para todos i y j .
 - (a) Use esta interpretación combinatoria para encontrar $M_{\lambda,\mu}(e, m)$ para $\lambda = (4, 2, 1)$ y $\mu = (3, 1^4)$.
 - (b) Use esta interpretación para demostrar que $M_{\lambda,\mu}(e, m) = M_{\mu,\lambda}(e, m)$.
 - (c) Describa una biyección entre la interpretación como tablas y la interpretación como matrices de $\{0, 1\}$.
8. Calcule $M_{\lambda,\mu}(h, m)$ para $\lambda = (2^2, 1^4)$ y $\mu = (3^2, 1^2)$.
9. Encuentre y pruebe una fórmula para $M_{\lambda,1^n}(h, m)$, para cualquier $\lambda \vdash n$.
10. Encuentre y pruebe una fórmula para $M_{\lambda,n}(h, m)$, para cualquier $\lambda \vdash n$.
11. Encuentre y pruebe una fórmula para $M_{\lambda,(n-1,1)}(h, m)$, para cualquier $\lambda \vdash n$.

12. Demuestre que para todo $k \geq 1$ y todas las particiones $\lambda, \mu \vdash k$, $M_{\lambda, \mu}(p, m)$ es el número de rellenos de Tablas de Young de λ UNSURE con enteros positivos, en los cuales las entradas de cada fila son constantes y cada entero i aparece μ_i veces.
13. Calcule $M_{\lambda, \mu}(p, m)$ para $\lambda = (2^2, 1^4)$ y $\mu = (3^2, 1^2)$.
14. Encuentre y pruebe una fórmula para $M_{\lambda, 1^n}(p, m)$, para cualquier $\lambda \vdash n$.
15. Encuentre y pruebe una fórmula para $M_{\lambda, n}(p, m)$, para cualquier $\lambda \vdash n$.
16. Encuentre y pruebe una fórmula para $M_{\lambda, (n-1, 1)}(p, m)$, para cualquier $\lambda \vdash n$.

5.3 Worksheet 3

1. Write the Schur function $s_{2,2}$ as a linear combination in the monomial, homogeneous, elementary, and power symmetric functions.
2. Prove the following for $\lambda, \mu \vdash k$:
 - (a) If $\mu >_{\text{lex}} \lambda$, then $K_{\lambda, \mu} = 0$
 - (b) $K_{\lambda, \lambda} = 1$.
3. Use Exercise 2 to prove that $\{s_\lambda : \lambda \vdash k\}$ is a basis for Λ^k .
4. How many semistandard tableaux (of any shape) have content k, k . I.e, this means k 1's and k 2's.
5. Prove that ω is an involution using the identity that for $n \geq 1$,

$$\sum_{i=0}^n (-1)^i e_i h_{n-i} = 0.$$

6. Use the Pieri rule to write $h_{3,2}$ in the Schur basis.
7. Use the Pieri rule to write $e_{3,2}$ in the Schur basis.
8. Write p_3 in the Schur basis.
9. Prove or disprove: for all $\lambda, \mu \vdash k$, we have $K_{\lambda, \mu} = K_{\mu, \lambda}$.
10. For each partition of $\lambda \vdash 3$, write h_λ as a linear combination of Schur functions. Make a conjecture about the coefficients you obtain.
11. Use the Jacobi-Trudi identity and induction to prove that for all n and k with $n \geq k \geq 0$, we have

$$s_{a, 1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}.$$

Español

1. Escriba la función de Schur $s_{2,2}$ como una combinación lineal de las funciones simétricas monomiales, homogéneas, elementales y de sumas de potencias.
2. Demuestre lo siguiente para $\lambda, \mu \vdash k$:

- (a) Si $\mu >_{\text{lex}} \lambda$, entonces $K_{\lambda,\mu} = 0$.
 (b) $K_{\lambda,\lambda} = 1$.
3. Use el Ejercicio 2 para demostrar que $\{s_\lambda : \lambda \vdash k\}$ es una base para Λ^k .
4. ¿Cuántos tableaux semiestándar (de cualquier forma) tienen contenido k, k ? Es decir, ¿cuántos tienen k entradas iguales a 1 y k entradas iguales a 2?
5. Demuestre que ω es una involución usando la identidad

$$\sum_{i=0}^n (-1)^i e_i h_{n-i} = 0,$$

válida para $n \geq 1$.

6. Use la regla de Pieri para escribir $h_{3,2}$ en la base de Schur.
7. Use la regla de Pieri para escribir $e_{3,2}$ en la base de Schur.
8. Escriba p_3 en la base de Schur.
9. Pruebe o dé un contraejemplo: para $\lambda, \mu \vdash k$, tenemos $K_{\lambda,\mu} = K_{\mu,\lambda}$.
10. Para cada partición $\lambda \vdash 3$, escriba h_λ como una combinación lineal de funciones de Schur. Formule una conjetura sobre los coeficientes obtenidos.
11. Use la identidad de Jacobi-Trudi y la inducción para demostrar que, para todos n y k con $n \geq k \geq 0$, se tiene

$$s_{a,1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}.$$

$$\mathbb{S}^{(2,2)} = \text{Span} \left\{ \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \right\}$$

5.4 Worksheet 4

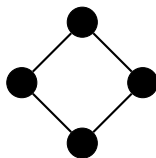
1. Explain why the chromatic symmetric function is symmetric.
2. Let G be a graph and e an edge in G . In graph theory the chromatic polynomial is computed using the deletion-contraction recurrence:

$$\chi_G(n) = \chi_{G-e}(n) - \chi_{G/e}(n),$$

where $G - e$ is the graph with the edge e removed and with the same vertices as G , and G/e is the graph with the edge e contracted to single vertex. If multiple edges result, you replace them with a single edge. Use this recurrence to compute the chromatic polynomial of the cycle graph with four vertices (the image is in Problem #3).

Recall that the chromatic polynomial of a tree, T , with k vertices is $\chi_T(n) = n(n-1)^{k-1}$

3. Take the cycle graph with four vertices and choose one edge.



- (a) What graph do you get when you delete this edge? Compute the chromatic symmetric function for the graph that you get in the power sum basis.
 - (b) What graph do you get when you contract this edge? Compute the chromatic symmetric function for the graph that you get in the power sum basis.
 - (c) Using this example, explain why the chromatic symmetric function does not satisfy the deletion-contraction recurrence.
4. A tree with k vertices can be defined as a connected graph with $k - 1$ edges.
 - (a) In our lecture we said that all trees have chromatic polynomial equal to $n(n-1)^{k-1}$. Use the binomial theorem,

$$(a + b)^m = \sum_{i=0}^m \binom{m}{i} a^i b^{m-i},$$

to expand this polynomial.

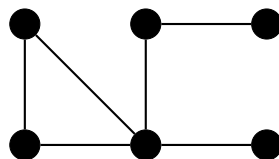
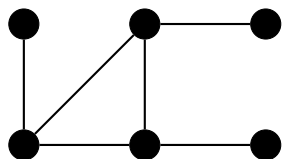
- (b) What is the coefficient of n^{k-1} ?

- (c) Use induction and the deletion-contraction recurrence to show that the coefficient of n^{k-1} in $\chi_G(n)$ is equal to the negative of the number of edges in G .
- (d) Suppose that G is a connected graph with k vertices and $\chi_G(n) = n(n-1)^{k-1}$, explain why G must be a tree.
5. Compute the chromatic symmetric function of the following graphs in the monomial basis.



6. Given X_G for some graph G in the power basis, how can you tell whether G is connected or not from the coefficients?
7. Show that if G and H satisfy $X_G = X_H$, then G and H must have the same number of edges.
8. Compute the chromatic symmetric function of the complete graph for $n \leq 5$ vertices in the elementary basis. Conjecture a formula for the chromatic symmetric function of the complete graph.
9. Let G be a graph with a leaf edge e ,
- Describe the graph that you get when you delete e .
 - Describe the graph that you get when you do a dot-contraction to e .
 - How are the graphs that you got in (a) and in (b) related?
 - What graph do you get when you leaf-contrast the edge e ?
 - How does this justify that we only apply deletion-near-contraction to internal edges?
10. Use the deletion-near-contraction relation to find the chromatic symmetric function of all trees with 5 vertices in the star basis.
11. Compute the chromatic symmetric function of the following two graphs in the star basis.

5. WORKSHEETS



Español

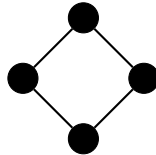
1. Explique por qué la función simétrica cromática es una función simétrica.
2. Sea G un grafo y e una arista en G . En teoría de grafos, el polinomio cromático se calcula usando la relación de eliminación-contracción:

$$\chi_G(n) = \chi_{G-e}(n) - \chi_{G/e}(n),$$

donde $G - e$ es el grafo con la arista e eliminada y con los mismos vértices que G , y G/e es el grafo que se obtiene al contraer la arista e a un solo vértice. Si resultan múltiples aristas, reemplácelas por una sola arista. Usando esta relación, calcule el polinomio cromático del grafo cíclico de cuatro vértices (la imagen del Problema #3).

Recuerde que el polinomio cromático de un árbol T con k vértices es $\chi_T(n) = n(n-1)^{k-1}$.

3. Tome el grafo cíclico con cuatro vértices y elija una arista.



- (a) ¿Qué grafo obtiene al eliminar esta arista? Calcule la función simétrica cromática del grafo obtenido en la base de sumas de potencias.
 - (b) ¿Qué grafo obtiene al contraer esta arista? Calcule la función simétrica cromática del grafo obtenido en la base de sumas de potencias.
 - (c) Usando este ejemplo, explique por qué la función simétrica cromática no satisface la relación de eliminación-contracción.
4. Un árbol con k vértices puede definirse como un grafo conexo con $k - 1$ aristas.
 - (a) En clase dijimos que todos los árboles tienen polinomio cromático igual a $n(n-1)^{k-1}$. Use el teorema del binomio,

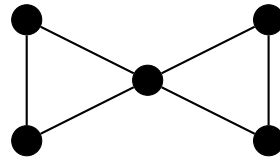
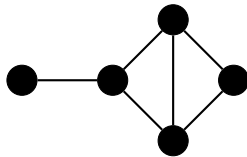
$$(a + b)^m = \sum_{i=0}^m \binom{m}{i} a^i b^{m-i},$$

para expandir este polinomio.

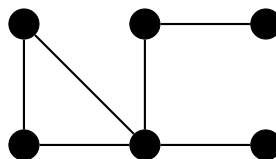
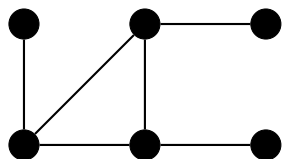
- (b) ¿Cuál es el coeficiente de n^{k-1} ?

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- (c) Use inducción y la relación de eliminación-contracción para demostrar que el coeficiente de n^{k-1} en $\chi_G(n)$ es igual al negativo del número de aristas de G .
 - (d) Suponga que G es un grafo conexo con k vértices y $\chi_G(n) = n(n-1)^{k-1}$; explique por qué G debe ser un árbol.
5. Calcule la función simétrica cromática de los siguientes grafos en la base monomial.



- 6. Dado X_G para algún grafo G en la base de sumas de potencias, ¿cómo puede determinar a partir de los coeficientes si G es conexo o no?
- 7. Demuestre que si G y H satisfacen $X_G = X_H$, entonces G y H deben tener el mismo número de aristas.
- 8. Calcule la función simétrica cromática del grafo completo para $n \leq 5$ vértices en la base elemental. Conjeture una fórmula para la función simétrica cromática del grafo completo.
- 9. Sea G un grafo con una arista hoja e :
 - (a) Describa el grafo que obtiene al eliminar e .
 - (b) Describa el grafo que obtiene al aplicar una contracción-punto (*dot-contraction*) a e .
 - (c) ¿Cómo se relacionan los grafos que obtuvo en (a) y (b)?
 - (d) ¿Qué grafo obtiene al aplicar una contracción-hoja (*leaf-contraction*) a la arista e ?
 - (e) ¿Cómo justifica esto que solo apliquemos la eliminación-cerca-de-contracción (*deletion-near-contraction*) a aristas internas?
- 10. Use la relación de eliminación-cerca-de-contracción, en inglés *deletion-near-contraction*, para encontrar la función simétrica cromática de todos los árboles con 5 vértices en la base de estrellas (*star basis*).
- 11. Calcule la función simétrica cromática de los siguientes dos grafos en la base de estrellas.



5.5 SageMath Worksheet

1. For every partition n , $n \leq 6$, compute the change of basis matrix from the m -basis to the s -basis.
 - (a) What ordering is Sage using?
 - (b) Is the matrix triangular?
 - (c) What is the determinant?
2. Choose several partitions λ and compute $s[\lambda].\text{expand}(4)$ for each. Make sure your examples include partitions with more and fewer than 4 parts.
 - (a) What does this function do?
 - (b) In each of your examples, use these expressions to write $s[\lambda].\text{expand}(4)$ as a sum of monomial symmetric functions.
 - (c) Look for patterns in the coefficients. What do you notice?
3. For $n \leq 7$ vertices, find the chromatic symmetric function of the following class of graphs: paths, cycles, and complete graphs.
 - (a) For each class, expand in the monomial, elementary, power, and Schur bases.
 - (b) Which basis has the simplest expression for each class of graphs?
 - (c) Does the chromatic symmetric function have all positive coefficients in any of the bases?
4. Compute the chromatic symmetric function for the star graph with n vertices, St_n , for $1 \leq n \leq 7$. Check the proposition about the expression X_{St_n} in the power symmetric function.
5. Generate all trees with $n = 7$ vertices and compute their chromatic symmetric functions. Are there two trees with the same function? Repeat with $n = 8$.
6. Expand the chromatic symmetric function of each tree on $n = 8$ vertices and write it in the elementary basis. Does the coefficient of $e_{4,2,1,1}$ distinguish these trees?
7. For every tree with 8 vertices, compute the following:
 - (a) degree sequence.
 - (b) diameter.

- (c) number of leaves.
- (d) the coefficient of $e_{n-1,1}$ and $p_{n-1,1}$.
- (e) Are there any correlations between the coefficients and items (a) - (c)?

Mini-Projects

A. **Question:** Which basis best encode trees?

Objective: Investigate how the chromatic symmetric function behaves in each of the bases.

- (a) Compute the chromatic symmetric function for each tree with $n = 8$ vertices.
- (b) For each tree, write its chromatic symmetric function in the monomial, elementary, power, and Schur basis.
- (c) Store the number of nonzero coefficients, largest coefficient, smallest coefficient.
- (d) Create a table such that list every tree and number of terms in each of the basis.
- (e) Which basis gives the sparsest expansion?
- (f) Is there a basis with all positive coefficients?
- (g) Is there anything that you can tell about the tree from the coefficients?

B. **Question:** What is the smallest and largest partitions in lexicographic order that appears with nonzero coefficient?

Objective: See if these partitions or their coefficients give information about the trees.

- (a) Compute the chromatic symmetric function for each tree with $n = 8$ vertices.
- (b) Expand the chromatic symmetric functions in the monomial basis.
- (c) For each tree write the smallest and largest partition in lexicographic largest and smallest partition.
- (d) Does it encode any information about the tree?
- (e) How does the coefficient change when you add a leaf to your tree?
- (f) Repeat the whole exercise for the other bases of symmetric functions.

Español

1. Para cada partición de n , con $n \leq 6$, calcule la matriz de cambio de base de la base m a la base s .
 - (a) ¿Qué orden está usando Sage?
 - (b) La matriz que resulta, ¿es triangular?
 - (c) ¿Cuál es su determinante?
2. Elija varias particiones λ y calcule $s[\lambda].\text{expand}(4)$ para cada una. Asegúrese de que sus ejemplos incluyan particiones con más y menos de 4 partes.
 - (a) ¿Qué hace esta función?
 - (b) En cada uno de sus ejemplos, use estas expresiones para escribir $s[\lambda].\text{expand}(4)$ como una suma de funciones simétricas monomiales.
 - (c) Busque patrones en los coeficientes. ¿Qué observa?
3. Para $n \leq 7$ vértices, encuentre la función simétrica cromática de las siguientes clases de grafos: caminos, ciclos y grafos completos.
 - (a) Para cada clase, expanda en las bases monomial, elemental, de potencias y de Schur.
 - (b) ¿Qué base tiene la expresión más sencilla para cada clase de grafos?
 - (c) ¿Tiene la función simétrica cromática todos sus coeficientes positivos en alguna de las bases?
4. Calcule la función simétrica cromática del grafo estrella con n vértices, St_n , para $1 \leq n \leq 7$. Compruebe la proposición sobre la expresión X_{St_n} en la función simétrica de potencias.
5. Genere todos los árboles con $n = 7$ vértices y calcule sus funciones simétricas cromáticas. Existen dos árboles con la misma función? Repita con $n = 8$.
6. Expanda la función simétrica cromática de cada árbol con $n = 8$ vértices y escríbala en la base elemental. ¿Distingue el coeficiente de $e_{4,2,1,1}$ a estos árboles?
7. Para cada árbol con 8 vértices, calcule lo siguiente:
 - (a) la sucesión de grados.
 - (b) el diámetro.

- (c) el número de hojas.
- (d) el coeficiente de $e_{n-1,1}$ y $p_{n-1,1}$.
- (e) ¿Hay alguna correlación entre los coeficientes y los elementos (a)–(c)?

Mini-Proyectos

A. **Pregunta:** ¿Qué base codifica mejor los árboles?

Objetivo: Investigar cómo se comporta la función simétrica cromática en cada una de las bases.

- (a) Calcule la función simétrica cromática de cada árbol con $n = 8$ vértices.
- (b) Para cada árbol, escriba su función simétrica cromática en las bases monomial, elemental, de potencias y de Schur.
- (c) Guarde el número de coeficientes no nulos, el coeficiente más grande y el coeficiente más pequeño.
- (d) Cree una tabla que enumere cada árbol y el número de términos en cada una de las bases.
- (e) ¿Qué base produce la expansión más esparza (*sparsest*)?
- (f) ¿Existe una base con todos sus coeficientes positivos?
- (g) ¿Hay algo que se pueda decir sobre el árbol a partir de los coeficientes?

B. **Pregunta:** ¿Cuáles son las particiones más pequeña y más grande en orden lexicográfico que aparecen con coeficiente no nulo?

Objetivo: Ver si estas particiones o sus coeficientes proporcionan información sobre los árboles.

- (a) Calcule la función simétrica cromática de cada árbol con $n = 8$ vértices.
- (b) Expanda las funciones simétricas cromáticas en la base monomial.
- (c) Para cada árbol, escriba la partición más pequeña y la más grande en orden lexicográfico.
- (d) ¿Proporciona esto alguna información sobre el árbol?
- (e) ¿Cómo cambia el coeficiente cuando añade una hoja a su árbol?
- (f) Repita todo el ejercicio para las otras bases de funciones simétricas.

Chapter 6

Solutions

6.1 Solutions to Worksheet 1

1. **Solution.** We use the action in which $\pi(x_i) = x_{\pi(i)}$. Write the three monomials of f as

$$A = x_1 x_2^2 x_4^4, \quad B = x_2^2 x_3^4 x_4, \quad C = x_1^4 x_2^2 x_3.$$

The variable x_2 is the only variable that has exponent 2 in every monomial. Therefore, any permutation that fixes f must fix 2.

The cycle $(1\ 4\ 3)$ fixes 2 and sends

$$A \mapsto B, \quad B \mapsto C, \quad C \mapsto A.$$

Thus $(1\ 4\ 3)$ fixes the sum f . Its inverse $(1\ 3\ 4)$ also fixes f , as does the identity.

There are no other possibilities. Indeed, after fixing 2, the image of A must be one of A , B , or C . Because the exponents 1, 0, and 4 on x_1, x_3, x_4 are distinct, each choice determines the permutation completely, and the three choices give precisely the permutations above. Hence

$$\{\pi \in S_4 : \pi(f) = f\} = \{\text{id}, (1\ 4\ 3), (1\ 3\ 4)\}.$$

2. **Solution.** Start with the monomial $x_1^7 x_3^2$ and take its alternating sum over S_3 :

$$f = \sum_{\pi \in S_3} \text{sgn}(\pi) \pi(x_1^7 x_3^2).$$

Since the exponents 7, 2, and 0 are distinct, the six monomials in this sum are distinct. Expanding gives

$$\begin{aligned} f &= x_1^7 x_3^2 - x_2^7 x_3^2 - x_1^2 x_3^7 - x_1^7 x_2^2 \\ &\quad + x_1^2 x_2^7 + x_2^2 x_3^7. \end{aligned}$$

In particular, $x_1^7 x_3^2$ occurs with coefficient 1.

To verify the required property, let $\sigma \in S_3$. Reindexing the sum with $\tau = \sigma\pi$ gives

$$\begin{aligned}\sigma(f) &= \sum_{\pi \in S_3} \text{sgn}(\pi) (\sigma\pi)(x_1^7 x_3^2) \\ &= \text{sgn}(\sigma) \sum_{\tau \in S_3} \text{sgn}(\tau) \tau(x_1^7 x_3^2) = \text{sgn}(\sigma)f.\end{aligned}$$

Alternative solution. The same polynomial can be written as the determinant

$$f = \begin{vmatrix} x_1^7 & x_2^7 & x_3^7 \\ 1 & 1 & 1 \\ x_1^2 & x_2^2 & x_3^2 \end{vmatrix}.$$

Permuting the variables permutes the columns of the determinant, so the determinant is multiplied by the sign of the permutation. Expanding the determinant also shows that the coefficient of $x_1^7 x_3^2$ is 1.

3. **Solution.** A monomial symmetric polynomial is the sum of all distinct monomials obtained by permuting the indicated exponents among the available variables. Therefore,

$$m_{3,2}(x_1, x_2) = x_1^3 x_2^2 + x_1^2 x_2^3,$$

and

$$\begin{aligned}m_{3,2}(x_1, x_2, x_3) &= x_1^3 x_2^2 + x_1^3 x_3^2 + x_2^3 x_1^2 \\ &\quad + x_2^3 x_3^2 + x_3^3 x_1^2 + x_3^3 x_2^2.\end{aligned}$$

The partition $(3, 3, 1, 1)$ has four positive parts, so it cannot be placed in only three variables. Hence

$$m_{3,3,1,1}(x_1, x_2, x_3) = 0.$$

With four variables, choose the two variables that receive exponent 3; the other two receive exponent 1. Thus

$$\begin{aligned}m_{3,3,1,1}(x_1, x_2, x_3, x_4) &= x_1^3 x_2^3 x_3 x_4 + x_1^3 x_3^3 x_2 x_4 + x_1^3 x_4^3 x_2 x_3 \\ &\quad + x_2^3 x_3^3 x_1 x_4 + x_2^3 x_4^3 x_1 x_3 + x_3^3 x_4^3 x_1 x_2.\end{aligned}$$

4. **Solution.** The coefficient of x^n in the product is obtained by the Cauchy product:

$$\begin{aligned} f(x)g(x) &= \sum_{n=0}^{\infty} \left(\sum_{\substack{i+j=n \\ i,j \geq 0}} j \right) x^n \\ &= \sum_{n=0}^{\infty} \left(\sum_{j=0}^n j \right) x^n = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n. \end{aligned}$$

Equivalently, as formal power series,

$$f(x) = \frac{1}{1-x}, \quad g(x) = \frac{x}{(1-x)^2},$$

so

$$f(x)g(x) = \frac{x}{(1-x)^3} = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n.$$

5. **Solution.** Distributing the two sums gives

$$\begin{aligned} fg &= \left(\sum_{i=1}^{\infty} x_i x_{i+1} \right) \left(\sum_{j=1}^{\infty} x_j \right) \\ &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} x_i x_{i+1} x_j. \end{aligned}$$

Separating the cases $j = i$, $j = i + 1$, and $j \notin \{i, i + 1\}$ gives the more explicit form

$$fg = \sum_{i=1}^{\infty} (x_i^2 x_{i+1} + x_i x_{i+1}^2) + \sum_{i=1}^{\infty} \sum_{\substack{j \geq 1 \\ j \neq i, i+1}} x_i x_{i+1} x_j.$$

The last double sum should not be simplified by discarding repetitions. For example, $x_i x_{i+1} x_{i+2}$ occurs twice: once using the adjacent pair $(i, i + 1)$ and once using the adjacent pair $(i + 1, i + 2)$.

6. **Solution.** By definition,

$$m_{2,1} = \sum_{i \neq j} x_i^2 x_j = \sum_{i < j} (x_i^2 x_j + x_i x_j^2).$$

On the other hand,

$$\begin{aligned} m_2 m_1 &= \left(\sum_i x_i^2 \right) \left(\sum_j x_j \right) \\ &= \sum_i x_i^3 + \sum_{i \neq j} x_i^2 x_j = m_3 + m_{2,1}. \end{aligned}$$

Therefore,

$$m_2 m_1 \neq m_{2,1}; \quad m_2 m_1 = m_3 + m_{2,1}.$$

7. **Solution.** In a product $m_3 m_2 m_1$, choose one variable for the exponent 3, one for the exponent 2, and one for the exponent 1. We classify the result according to which choices are equal.

- ◇ If all three choices are the same, we obtain m_6 .
- ◇ If only the choices for exponents 3 and 2 are the same, we obtain $m_{5,1}$.
- ◇ If only the choices for exponents 3 and 1 are the same, we obtain $m_{4,2}$.
- ◇ If only the choices for exponents 2 and 1 are the same, we obtain $m_{3,3}$. Each monomial of this type occurs twice, because either of its two variables can come from the m_3 factor.
- ◇ If all three choices are distinct, we obtain $m_{3,2,1}$.

Consequently,

$$m_3 m_2 m_1 = m_6 + m_{5,1} + m_{4,2} + 2m_{3,3} + m_{3,2,1}.$$

8. **Solution.** Recall that

$$m_{1^r} = \sum_{\substack{A \subseteq \{1, 2, \dots\} \\ |A|=r}} \prod_{i \in A} x_i.$$

Fix one monomial of type $(2, 1^{a+b})$. It uses $a + b + 1$ distinct variables: one variable is squared and the other $a + b$ variables occur to the first power.

To obtain this monomial from $m_{1^{a+1}} m_{1^{b+1}}$, the squared variable must be selected in both factors. Of the remaining $a + b$ variables, choose a to join it in the first factor; the remaining b must be selected in the second factor. There are

$$\binom{a+b}{a} = \binom{a+b}{b}$$

ways to do this. By symmetry every monomial in $m_{2,1^{a+b}}$ has the same coefficient. Therefore the requested coefficient is

$$\binom{a+b}{a}.$$

9. Solution.

(a) In forming the product, we choose either 1 or x_i from each factor. A term obtained by choosing r distinct variables is a monomial of type (1^r) . Hence

$$\prod_{i=1}^{\infty} (1 + x_i) = \sum_{r=0}^{\infty} m_{1^r} = 1 + m_1 + m_{1,1} + m_{1,1,1} + \cdots.$$

(b) The same argument applies, but choosing $-x_i$ from r factors contributes the sign $(-1)^r$. Therefore,

$$\prod_{i=1}^{\infty} (1 - x_i) = \sum_{r=0}^{\infty} (-1)^r m_{1^r} = 1 - m_1 + m_{1,1} - m_{1,1,1} + \cdots.$$

10. Solution. For each variable, use the geometric series

$$1 + x_i + x_i^2 + \cdots = \frac{1}{1 - x_i}.$$

When the products are multiplied out, every monomial $x_1^{a_1} x_2^{a_2} \cdots$ of finite total degree occurs exactly once. Grouping monomials whose positive exponents form the same partition λ gives

$$\prod_{i=1}^{\infty} \frac{1}{1 - x_i} = 1 + \sum_{n=1}^{\infty} \sum_{\lambda \vdash n} m_{\lambda}.$$

The initial 1 corresponds to choosing 1 from every factor, or equivalently to the empty partition. Since the worksheet specifies partitions of positive integers, the requested sum is therefore

$$\sum_{n=1}^{\infty} \sum_{\lambda \vdash n} m_{\lambda} = \prod_{i=1}^{\infty} \frac{1}{1 - x_i} - 1.$$

If the empty partition is included and its monomial symmetric function is defined to be 1, then the answer is simply $\prod_{i=1}^{\infty} (1 - x_i)^{-1}$.

6.2 Solutions to Worksheet 2

1. **Solution.** For the change of basis from e to m , the number $M_{\lambda,\mu}(e, m)$ counts fillings of the rows of shape λ with content μ such that every row is *strictly increasing*. There is no condition down the columns. Rows in different positions are regarded as different, even when they have the same length.

Here the shape $(2^2, 1^4) = (2, 2, 1, 1, 1, 1)$ has two rows of length 2 and four one-cell rows. Call the long rows A and B . The content $(3^2, 1^2) = (3, 3, 1, 1)$ means that the complete filling must contain three 1's, three 2's, one 3, and one 4. Because a long row is strictly increasing, it cannot contain two equal entries.

We count according to where the exceptional entries 3 and 4 occur.

- (a) Suppose neither 3 nor 4 is in a long row. Then both long rows must be $[1, 2]$. The four one-cell rows contain 1, 2, 3, 4 in any order, giving

$$4! = 24$$

fillings.

- (b) Suppose exactly one of 3, 4 is in a long row. Choose which exceptional entry is used (2 choices), choose whether it is in A or B (2 choices), and choose whether its companion is 1 or 2 (2 choices). The other long row is then forced to be $[1, 2]$. The four one-cell rows contain two equal entries and two other entries, so they can be filled in

$$\frac{4!}{2!} = 12$$

ways. This case contributes

$$2 \cdot 2 \cdot 2 \cdot 12 = 96.$$

- (c) Suppose both 3 and 4 occur in the long rows. There are three possibilities.

- i. If 3 and 4 are together, choose their long row in 2 ways. The other long row must be $[1, 2]$, and the one-cell rows contain 1, 1, 2, 2. This gives

$$2 \cdot \frac{4!}{2!2!} = 12.$$

- ii. If 3 and 4 are in different long rows and have the same companion, choose that companion in 2 ways and choose which long row contains 3 in 2 ways. The one-cell rows contain three copies of one of 1, 2 and one copy of the other, so this gives

$$2 \cdot 2 \cdot \frac{4!}{3!} = 16.$$

- iii. If 3 and 4 are in different long rows and have different companions, choose which long row contains 3 in 2 ways and choose which exceptional entry is paired with 1 in 2 ways. The one-cell rows contain 1, 1, 2, 2, so this gives

$$2 \cdot 2 \cdot \frac{4!}{2!2!} = 24.$$

Thus this case contributes $12 + 16 + 24 = 52$.

Adding the three cases, we obtain

$$M_{(2^2, 1^4), (3^2, 1^2)}(e, m) = 24 + 96 + 52 = 172.$$

2. **Solution.** We use row-strict tableaux of shape λ and content 1^n . The content $1^n = (1, 1, \dots, 1)$ means that the entries $1, 2, \dots, n$ each appear exactly once.

First choose which λ_1 entries go in the first row, then which λ_2 of the remaining entries go in the second row, and so on. Once the entries of a row have been chosen, there is only one legal way to place them: in increasing order. Therefore

$$M_{\lambda, 1^n}(e, m) = \binom{n}{\lambda_1, \dots, \lambda_\ell} = \frac{n!}{\lambda_1! \cdots \lambda_\ell!}.$$

3. **Solution.** The content (n) means that every one of the n cells must contain 1. A strictly increasing row cannot contain two 1's. Therefore every row must have length 1, which is exactly the shape $\lambda = 1^n$. For this shape there is one filling: put 1 in every cell. Hence

$$M_{\lambda, n}(e, m) = \begin{cases} 1, & \lambda = 1^n, \\ 0, & \text{otherwise.} \end{cases}$$

4. **Solution.** Assume $n \geq 2$. The content $(n-1, 1)$ means that the tableau has $n-1$ entries equal to 1 and one entry equal to 2.

Since rows are strictly increasing, no row can contain two 1's. Thus every row has length 1, except possibly one row of length 2. Consequently, the only possible shapes are 1^n and $(2, 1^{n-2})$.

For shape 1^n , choose which of the n one-cell rows contains the unique 2; all other rows contain 1. This gives n fillings. For shape $(2, 1^{n-2})$, the long row is forced to be

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$[1, 2]$, and every one-cell row is forced to contain 1, giving one filling. Therefore

$$M_{\lambda, (n-1, 1)}(e, m) = \begin{cases} n, & \lambda = 1^n, \\ 1, & \lambda = (2, 1^{n-2}), \\ 0, & \text{otherwise.} \end{cases}$$

5. **Solution.** The statement is false. Take

$$\lambda = (4, 1, 1), \quad \mu = (3, 3).$$

Both are partitions of 6, and $\lambda >_{\text{lex}} \mu$ because $4 > 3$. Their conjugates are

$$\lambda' = (3, 1, 1, 1), \quad \mu' = (2, 2, 2).$$

Now $\mu' <_{\text{lex}} \lambda'$, rather than $\mu' >_{\text{lex}} \lambda'$. This is a counterexample.

6. **Solution.** Let

$$E(t) = \prod_{i \geq 1} (1 + x_i t) = \sum_{r \geq 0} e_r t^r$$

and let ω be a primitive cube root of unity, so $1 + \omega + \omega^2 = 0$. Since

$$1 + x + x^2 = (1 - \omega x)(1 - \omega^2 x),$$

we obtain

$$\prod_{i \geq 1} (1 + x_i + x_i^2) = E(-\omega)E(-\omega^2) = \left(\sum_{r \geq 0} (-\omega)^r e_r \right) \left(\sum_{s \geq 0} (-\omega^2)^s e_s \right).$$

This expression has integer coefficients because the two factors are complex conjugates. For example, its first homogeneous pieces are

$$1 + e_1 + (e_1^2 - e_2) + (e_1 e_2 - 2e_3) + \cdots.$$

7. **Solution.**

- (a) Only the first five rows and first three columns can be nonzero. The row sums are $(3, 1, 1, 1, 1)$ and the column sums are $(4, 2, 1)$. The row of sum 3 must be $(1, 1, 1)$. After removing it, the four remaining rows each contain one 1, and the remaining column sums are $(3, 1, 0)$. Choose which three of the four rows place their 1 in the first column; the remaining row uses the second column. Thus

$$M_{(4, 2, 1), (3, 1^4)}(e, m) = \binom{4}{3} = 4.$$

- (b) Transposition is a bijection. A matrix with row sums μ_i and column sums λ_j is sent by transposition to a matrix with row sums λ_j and column sums μ_i . Therefore

$$M_{\lambda,\mu}(e, m) = M_{\mu,\lambda}(e, m).$$

- (c) In the tableaux interpretation, row j of the Young diagram of λ is filled with distinct positive integers, written in increasing order, and integer i occurs μ_i times. Define $a_{ij} = 1$ when integer i occurs in tableau row j , and define $a_{ij} = 0$ otherwise. Distinctness within a tableau row makes this a $\{0, 1\}$ -matrix. Its row sum is μ_i and its column sum is λ_j .

Conversely, from such a matrix, fill tableau row j with the indices i for which $a_{ij} = 1$, in increasing order. These constructions are inverse to one another, so they give the required bijection.

8. **Solution.** For the change of basis from h to m , the number $M_{\lambda,\mu}(h, m)$ counts fillings of the rows of shape λ with content μ such that every row is *weakly increasing*. Again, there is no condition down the columns, and equal-length rows in different positions are distinguished.

We have two long rows A, B of length 2 and four one-cell rows. We must place three 1's, three 2's, one 3, and one 4. The main difference from Exercise 1 is that repeated entries are now allowed in a long row: for example, $[1, 1]$ and $[2, 2]$ are legal.

We count according to how many of 3, 4 occur in the long rows.

- (a) Suppose neither 3 nor 4 is in a long row. Each of A, B is one of

$$[1, 1], \quad [1, 2], \quad [2, 2].$$

The pair $([1, 1], [1, 1])$ is impossible because it uses four 1's, and $([2, 2], [2, 2])$ is impossible for the same reason with 2's. The legal possibilities contribute as follows:

types of A, B	orders	entries in one-cell rows	count
$\{[1, 1], [1, 2]\}$	2	2, 2, 3, 4	$2 \cdot 4!/2! = 24$
$\{[1, 1], [2, 2]\}$	2	1, 2, 3, 4	$2 \cdot 4! = 48$
$\{[1, 2], [1, 2]\}$	1	1, 2, 3, 4	$4! = 24$
$\{[1, 2], [2, 2]\}$	2	1, 1, 3, 4	$2 \cdot 4!/2! = 24$

Thus this case contributes 120.

(b) Suppose exactly one of 3, 4 is in a long row. Choose that exceptional entry in 2 ways and choose its long row in 2 ways. Its companion is either 1 or 2.

If its companion is 1, the other long row may be $[1, 1]$, $[1, 2]$, or $[2, 2]$. The corresponding one-cell rows can be filled in

$$\frac{4!}{3!} = 4, \text{quad} \frac{4!}{2!} = 12, \text{quad} \frac{4!}{2!} = 12$$

ways, respectively. This gives $4 + 12 + 12 = 28$ fillings. If the companion is 2, the same argument with 1 and 2 interchanged gives another 28. Therefore this case contributes

$$2 \cdot 2 \cdot (28 + 28) = 224.$$

(c) Suppose both 3 and 4 are in the long rows.

i. They may be together in one long row, which can be chosen in 2 ways. The other long row may be $[1, 1]$, $[1, 2]$, or $[2, 2]$. The one-cell rows can then be filled in 4, 6, or 4 ways, respectively. This gives

$$2(4 + 6 + 4) = 28.$$

ii. They may be in different long rows. There are 2 ways to decide which long row contains 3. If both companions are 1, the one-cell rows can be filled in 4 ways; if both are 2, there are 4 ways; and if the companions are different, there are 2 choices for their assignment and 6 fillings of the one-cell rows. This gives

$$2(4 + 4 + 2 \cdot 6) = 40.$$

Thus this case contributes $28 + 40 = 68$.

Adding all three cases gives

$$M_{(2^2, 1^4), (3^2, 1^2)}(h, m) = 120 + 224 + 68 = 412.$$

9. **Solution.** We count weakly increasing row fillings of shape λ and content 1^n . Since the entries $1, 2, \dots, n$ are all distinct, choose which λ_1 entries go in the first row, which λ_2 of the remaining entries go in the second row, and so forth. After the entries of a row have been chosen, their weakly increasing order is forced; because the entries are distinct, it is actually strictly increasing. Thus

$$M_{\lambda, 1^n}(h, m) = \frac{n!}{\lambda_1! \cdots \lambda_\ell!}.$$

10. **Solution.** The content (n) means that every cell contains 1. This is allowed because a row only has to be weakly increasing, so repeated entries are permitted. Every row is therefore forced, and there is exactly one tableau of every shape $\lambda \vdash n$. Thus

$$M_{\lambda,n}(h, m) = 1 \quad \text{for every } \lambda \vdash n.$$

11. **Solution.** The content $(n-1, 1)$ consists of $n-1$ copies of 1 and one copy of 2. Choose the row that will contain the unique 2. In that row, weak increase forces the 2 to be the final entry, with all earlier entries equal to 1. Every other row consists entirely of 1's.

Thus each row gives exactly one tableau, depending on whether that row is chosen to contain the 2. Since λ has $\ell(\lambda)$ rows, including rows of equal length in different positions, we obtain

$$M_{\lambda,(n-1,1)}(h, m) = \ell(\lambda),$$

where $\ell(\lambda)$ is the number of parts of λ .

12. **Solution.** Write

$$p_\lambda = p_{\lambda_1} \cdots p_{\lambda_\ell}, \quad p_{\lambda_j} = \sum_{i \geq 1} x_i^{\lambda_j}.$$

For each row j , choosing the term $x_i^{\lambda_j}$ from p_{λ_j} means placing the entry i in all λ_j cells of that row. Therefore the relevant tableaux are exactly the fillings in which *every row is constant*. There is no condition relating one row to another.

After making one choice for every row, the exponent of x_i is the total number of cells filled with i . Hence the resulting monomial has exponent sequence μ exactly when the filling has content μ . We have therefore proved the combinatorial interpretation

$$M_{\lambda,\mu}(p, m) = \#\{\text{constant-row fillings of shape } \lambda \text{ and content } \mu\}.$$

13. **Solution.** We count constant-row tableaux of shape $(2, 2, 1, 1, 1, 1)$ and content $(3, 3, 1, 1)$. Thus we need three 1's, three 2's, one 3, and one 4.

The two long rows cannot be filled with 3 or 4, because a constant row of length 2 would use two copies of that entry, but the content provides only one copy of each. They also cannot both be filled with 1, since that would require four 1's, and they cannot both be filled with 2. Therefore one long row must be $[1, 1]$ and the other must be $[2, 2]$.

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There are 2 ways to decide which long row gets the 1's. After that choice, the four one-cell rows must contain one each of 1, 2, 3, 4, and these four entries can be assigned to the four rows in $4!$ ways. Hence

$$M_{(2^2,1^4),(3^2,1^2)}(p, m) = 2 \cdot 4! = 48.$$

14. **Solution.** The content 1^n uses each of the entries $1, 2, \dots, n$ exactly once. A constant row of length greater than 1 would repeat its entry, so such a row is impossible. Consequently every row must have length 1, which means $\lambda = 1^n$.

For this shape, assign the n distinct entries to the n one-cell rows. The first row has n choices, the second has $n - 1$, and so on, for a total of $n!$ tableaux. Therefore

$$M_{\lambda, 1^n}(p, m) = \begin{cases} n!, & \lambda = 1^n, \\ 0, & \text{otherwise.} \end{cases}$$

15. **Solution.** The content (n) means that all n cells contain 1. Every row is then constant automatically, and there are no choices to make. Thus there is exactly one such tableau for every shape $\lambda \vdash n$, and

$$M_{\lambda, n}(p, m) = 1 \quad \text{for every } \lambda \vdash n.$$

16. **Solution.** The content $(n - 1, 1)$ consists of $n - 1$ copies of 1 and one copy of 2. Because rows must be constant, the row containing 2 cannot have length greater than 1: otherwise it would contain at least two 2's. Thus the unique 2 must occupy a one-cell row. After choosing that row, every other row is forced to contain only 1's. There is one choice for each part of λ equal to 1. If $m_1(\lambda)$ denotes the number of such parts, then

$$M_{\lambda, (n-1, 1)}(p, m) = m_1(\lambda),$$

where $m_1(\lambda)$ denotes the multiplicity of the part 1 in λ .

6.3 Solutions to Worksheet 3

1. **Solution.** The semistandard tableaux of shape $(2, 2)$ give

$$s_{2,2} = m_{2,2} + m_{2,1,1} + 2m_{1,1,1,1}.$$

For example, the coefficient 2 is the number of standard Young tableaux of shape $(2, 2)$.

The Jacobi–Trudi identity gives

$$s_{2,2} = h_2^2 - h_3 h_1 = h_{2,2} - h_{3,1}.$$

Since $(2, 2)$ is self-conjugate and the involution ω exchanges h_r with e_r , we also have

$$s_{2,2} = e_2^2 - e_3 e_1 = e_{2,2} - e_{3,1}.$$

Finally, using the character values for the representation indexed by $(2, 2)$, or converting the preceding formulas with Newton's identities, gives

$$s_{2,2} = \frac{1}{12}p_{1,1,1,1} + \frac{1}{4}p_{2,2} - \frac{1}{3}p_{3,1}.$$

2. **Solution.**

- (a) Suppose a semistandard tableau of shape λ and content μ exists. Because columns strictly increase, an entry in row r is at least r . Therefore all entries among $1, 2, \dots, r$ must lie in the first r rows. It follows that

$$\mu_1 + \dots + \mu_r \leq \lambda_1 + \dots + \lambda_r \quad \text{for every } r.$$

If $\mu >_{\text{lex}} \lambda$, let r be the first index at which they differ. Then $\mu_i = \lambda_i$ for $i < r$ and $\mu_r > \lambda_r$, contradicting the displayed inequality. Hence

$$\mu >_{\text{lex}} \lambda \implies K_{\lambda, \mu} = 0.$$

- (b) In a tableau of shape and content λ , all λ_1 copies of 1 must lie in the first row, so that row consists entirely of 1's. After removing it, all λ_2 copies of 2 must fill the second row. Continuing in this way shows that row i must consist entirely of i 's. This tableau is semistandard and is the only possibility. Thus

$$K_{\lambda, \lambda} = 1.$$

3. **Solution.** The monomial expansion of a Schur function is

$$s_\lambda = \sum_{\mu \vdash k} K_{\lambda, \mu} m_\mu.$$

Order the partitions of k lexicographically. Exercise 2 says that this change-of-basis matrix is triangular, and its diagonal entries are all $K_{\lambda, \lambda} = 1$. Its determinant is therefore 1, so it is invertible. Since the monomial functions $\{m_\mu : \mu \vdash k\}$ form a basis of Λ^k , the same is true of

$$\{s_\lambda : \lambda \vdash k\}.$$

4. **Solution.** Only the entries 1 and 2 are available. Since columns strictly increase, a tableau can have at most two rows. Its shape must be

$$(2k - r, r) \quad \text{for some } 0 \leq r \leq k.$$

In the r columns of height two, the top entry must be 1 and the bottom entry must be 2. The remaining cells are all in the first row; weak increase forces the remaining $k - r$ copies of 1 to come before the remaining $k - r$ copies of 2. Thus there is exactly one tableau for each value of r . The answer is

$$k + 1.$$

5. **Solution.** Let

$$E(t) = \sum_{n \geq 0} e_n t^n, \quad H(t) = \sum_{n \geq 0} h_n t^n.$$

The given identity is exactly the statement

$$E(-t)H(t) = 1.$$

Suppose ω is defined by $\omega(e_n) = h_n$ and extended as an algebra homomorphism. Apply ω to the displayed identity:

$$H(-t)\omega(H(t)) = 1.$$

Replacing t by $-t$ in the original identity also gives $E(t)H(-t) = 1$. An inverse is unique, so

$$\omega(H(t)) = E(t).$$

Comparing coefficients yields $\omega(h_n) = e_n$. Thus ω^2 fixes every e_n (and every h_n). Since the e_n generate the ring of symmetric functions,

$$\omega^2 = \text{id}.$$

6. **Solution.** Since $h_r = s_r$, apply the Pieri rule to $s_3 h_2$. Adding a horizontal strip of two boxes to the row (3) produces exactly (5), (4, 1), and (3, 2). Hence

$$h_{3,2} = s_5 + s_{4,1} + s_{3,2}.$$

7. **Solution.** Apply ω to the answer in Exercise 6. The involution exchanges h and e and sends s_λ to $s_{\lambda'}$. The conjugates of (5), (4, 1), and (3, 2) are respectively (1^5) , $(2, 1^3)$, and $(2, 2, 1)$. Therefore

$$e_{3,2} = s_{1^5} + s_{2,1^3} + s_{2,2,1}.$$

8. **Solution.** Newton's identities give

$$p_3 = 3h_3 - 3h_2 h_1 + h_1^3.$$

By the Pieri rule,

$$h_3 = s_3, \quad h_2 h_1 = s_3 + s_{2,1}, \quad h_1^3 = s_3 + 2s_{2,1} + s_{1,1,1}.$$

Substitution gives

$$p_3 = s_3 - s_{2,1} + s_{1,1,1}.$$

9. **Solution.** The statement is false. Let $\lambda = (2)$ and $\mu = (1, 1)$. There is one semistandard tableau of shape (2) and content (1, 1), namely a row containing 1, 2. Thus $K_{(2),(1,1)} = 1$. A column of shape (1, 1) cannot be filled with two copies of 1 because columns must strictly increase, so $K_{(1,1),(2)} = 0$. Hence the two Kostka numbers need not be equal.
10. **Solution.** The partitions of 3 give

$$\begin{aligned} h_3 &= s_3, \\ h_{2,1} &= s_3 + s_{2,1}, \\ h_{1,1,1} &= s_3 + 2s_{2,1} + s_{1,1,1}. \end{aligned}$$

The pattern is Young's rule:

$$h_\lambda = \sum_{\mu \vdash 3} K_{\mu,\lambda} s_\mu.$$

In general, the coefficient of s_μ in h_λ is the number of semistandard tableaux of shape μ and content λ .

11. **Solution.** For $a \geq 1$ and $b \geq 0$, the Jacobi–Trudi determinant for the hook $(a, 1^b)$ is

$$s_{a,1^b} = \begin{vmatrix} h_a & h_{a+1} & h_{a+2} & \cdots & h_{a+b} \\ 1 & h_1 & h_2 & \cdots & h_b \\ 0 & 1 & h_1 & \cdots & h_{b-1} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & h_1 \end{vmatrix}.$$

Expand along the first row. The minor corresponding to h_{a+i} is the Jacobi–Trudi determinant for e_{b-i} ; the cofactor sign is $(-1)^i$. Therefore

$$s_{a,1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}.$$

This also proves the formula inductively: the case $b = 0$ is $s_a = h_a$, and increasing b adds one more cofactor to the same Jacobi–Trudi expansion.

6.4 Solutions to Worksheet 4

1. **Solution.** Recall that

$$X_G = \sum_{\kappa} \prod_{v \in V(G)} x_{\kappa(v)},$$

where the sum is over proper colorings by positive integers. Let σ be any permutation of the positive integers. Replacing every color i by $\sigma(i)$ is a bijection from proper colorings to proper colorings. It permutes the variables in every monomial but leaves the whole sum unchanged. Thus X_G is invariant under every permutation of the variables, so it is symmetric.

2. **Solution.** Let e be any edge of C_4 . Deleting e produces the path P_4 , which is a tree on four vertices, while contracting e produces the triangle C_3 . Therefore

$$\chi_{C_4}(n) = n(n-1)^3 - n(n-1)(n-2).$$

Simplifying,

$$\chi_{C_4}(n) = n(n-1)(n^2 - 3n + 3) = n^4 - 4n^3 + 6n^2 - 3n.$$

3. **Solution.** We use Stanley's power-sum formula

$$X_G = \sum_{A \subseteq E(G)} (-1)^{|A|} p_{\lambda(A)},$$

where $\lambda(A)$ is the partition of the component sizes of the spanning subgraph $(V(G), A)$.

- (a) Deleting the chosen edge gives P_4 . Among its three edges, there are three one-edge subsets, two adjacent two-edge subsets, one disjoint two-edge subset, and one three-edge subset. Hence

$$X_{P_4} = p_{1,1,1,1} - 3p_{2,1,1} + 2p_{3,1} + p_{2,2} - p_4.$$

- (b) Contracting the edge gives $C_3 = K_3$. The subset formula gives

$$X_{C_3} = p_{1,1,1} - 3p_{2,1} + 2p_3.$$

- (c) The two terms in a deletion–contraction formula do not even have the same degree: X_{P_4} has degree 4, while X_{C_3} has degree 3. More explicitly,

$$X_{C_4} = p_{1,1,1,1} - 4p_{2,1,1} + 4p_{3,1} + 2p_{2,2} - 3p_4,$$

which is not $X_{P_4} - X_{C_3}$. Thus the ordinary deletion–contraction relation does not hold for the chromatic symmetric function.

4. Solution.

- (a) By the binomial theorem,

$$\begin{aligned} n(n-1)^{k-1} &= n \sum_{i=0}^{k-1} \binom{k-1}{i} n^i (-1)^{k-1-i} \\ &= \boxed{\sum_{j=1}^k (-1)^{k-j} \binom{k-1}{j-1} n^j}. \end{aligned}$$

- (b) Taking $j = k - 1$, the coefficient is

$$\boxed{-\binom{k-1}{k-2} = -(k-1)}.$$

- (c) For a simple graph,

$$\boxed{[n^{k-1}] \chi_G(n) = -|E(G)|}.$$

We prove this by induction on the number of edges. For the graph with no edges, $\chi_G(n) = n^k$, and both sides are 0. Choose an edge e . By induction, the coefficient of n^{k-1} in $\chi_{G-e}(n)$ is $-(|E(G)| - 1)$. The contracted graph has $k - 1$ vertices, so its chromatic polynomial is monic of degree $k - 1$. Consequently, deletion–contraction gives

$$[n^{k-1}] \chi_G(n) = -(|E(G)| - 1) - 1 = -|E(G)|.$$

- (d) If $\chi_G(n) = n(n-1)^{k-1}$, part (b) says that the coefficient of n^{k-1} is $-(k-1)$. By part (c), G has $k-1$ edges. Since G is connected, a connected graph on k vertices with $k-1$ edges is a tree.

5. **Solution.** Call the graphs G_1 and G_2 in their displayed order. In either graph there are four independent pairs. Thus a coloring of type $(2, 1, 1, 1)$ is obtained by choosing one of those pairs and then assigning the three singleton colors in $3!$ ways, giving coefficient $4 \cdot 3! = 24$.

There are two ways in either graph to choose two disjoint independent pairs. Since the two pair-colors can be interchanged, the coefficient of $m_{2,2,1}$ is $2 \cdot 2! = 4$. No independent set has size 3, and a coloring with all colors distinct contributes $5! = 120$. Therefore the two graphs have the same answer:

$$X_{G_1} = X_{G_2} = 4m_{2,2,1} + 24m_{2,1,1,1} + 120m_{1,1,1,1,1}.$$

6. **Solution.** In the power expansion, the coefficient of p_n is

$$\sum_{\substack{A \subseteq E(G) \\ (V(G), A) \text{ connected}}} (-1)^{|A|}.$$

It is zero when G is disconnected, because no edge subset can connect all the vertices. When G is connected, the signed sum is nonzero; up to the sign $(-1)^{n-1}$, it counts acyclic orientations having a unique sink at any fixed vertex. At least one such orientation is obtained by ordering vertices by their distance from the chosen sink and directing edges toward smaller distance. Consequently,

$$G \text{ is connected if and only if } [p_n]X_G \neq 0.$$

More generally, the smallest number of parts among partitions indexing nonzero power-sum terms is the number of connected components.

7. **Solution.** The term $p_{2,1^{n-2}}$ in the subset expansion can arise only from a subset containing exactly one edge. Every edge contributes $-p_{2,1^{n-2}}$. Hence

$$[p_{2,1^{n-2}}]X_G = -|E(G)|.$$

If $X_G = X_H$, these coefficients agree, so G and H have the same number of edges.

8. **Solution.** In a proper coloring of K_n , all vertices receive distinct colors. For each set of n colors there are $n!$ assignments to the vertices, so

$$X_{K_n} = n!e_n.$$

In particular,

$$X_{K_1} = e_1, \quad X_{K_2} = 2e_2, \quad X_{K_3} = 6e_3, \quad X_{K_4} = 24e_4, \quad X_{K_5} = 120e_5.$$

9. **Solution.** Let $e = uv$, where u is the leaf and v is its neighbor.

- (a) Deleting e leaves u isolated. Thus $G \setminus e = (G - u) \sqcup K_1$.
- (b) Dot-contraction contracts u and v and then adds an isolated vertex. Contracting a leaf edge simply removes the leaf, so the result is again $(G - u) \sqcup K_1$.
- (c) The graphs in (a) and (b) are isomorphic.
- (d) Leaf-contraction contracts u and v and then attaches a new leaf to the merged vertex. This recreates a graph isomorphic to G .
- (e) The deletion-near-contraction relation is

$$X_G = X_{G \setminus e} - X_{\text{dot}(G,e)} + X_{\text{leaf}(G,e)}.$$

For a leaf edge, the first two terms cancel and the last term is X_G , so the relation reduces to the tautology $X_G = X_G$. It makes no progress. This is why the recursive algorithm uses internal edges.

10. **Solution.** Write $st_r = X_{St_r}$ and $st_\lambda = \prod_i st_{\lambda_i}$. There are three trees on five vertices: St_5 , the tree T_5 with degree sequence $(3, 2, 1, 1, 1)$, and the path P_5 .

For T_5 , apply DNC to the unique internal edge adjacent to the degree-3 vertex. The deletion, dot-contraction, and leaf-contraction terms are respectively star forests of types $(3, 2)$, $(4, 1)$, and (5) . Thus

$$X_{T_5} = st_5 - st_{4,1} + st_{3,2}.$$

Similarly, applying DNC to the middle edge of P_4 gives

$$X_{P_4} = st_{2,2} - st_{3,1} + st_4.$$

Applying DNC to an internal edge of P_5 then gives

$$\begin{aligned} X_{P_5} &= st_{3,2} - X_{P_4} st_1 + X_{T_5} \\ &= st_5 - 2st_{4,1} + 2st_{3,2} + st_{3,1,1} - st_{2,2,1}. \end{aligned}$$

Finally, by definition,

$$X_{St_5} = st_5.$$

11. **Solution.** Call the displayed graphs H_1 and H_2 . The subset formula gives the same power-sum expansion for both:

$$\begin{aligned} X_{H_i} = & p_1^6 - 6p_{2,1^4} + 6p_{2,2,1,1} - p_{2,2,2} + 8p_{3,1,1,1} \\ & - 6p_{3,2,1} - 9p_{4,1,1} + 3p_{4,2} + 6p_{5,1} - 2p_6. \end{aligned}$$

To convert to the star basis, one may use

$$p_n = \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} \text{st}_{r+1, 1^{n-1-r}}$$

and multiply this identity for the parts of each power-sum index. After collecting terms,

$$\begin{aligned} X_{H_1} = X_{H_2} = & 2\text{st}_6 - 4\text{st}_{5,1} + 3\text{st}_{4,2} + 2\text{st}_{4,1,1} \\ & - 3\text{st}_{3,2,1} + \text{st}_{2,2,2}. \end{aligned}$$

Thus these two nonisomorphic graphs have the same chromatic symmetric function.

6.5 SageMath Solutions

The computations below were checked with SageMath 10.8. We begin every session by creating the four bases that occur in the worksheet.

```
Sym = SymmetricFunctions(QQ)
m = Sym.monomial()
e = Sym.elementary()
p = Sym.powersum()
s = Sym.schur()
```

Calling, for example, `e(X)` converts the symmetric function `X` to the elementary basis.

1. **Solution.** The following code computes all the requested matrices.

```
for n in range(1, 7):
    order = list(Partitions(n))
    A = m.transition_matrix(s, n)
    upper = all(A[i,j] == 0
                for i in range(A.nrows())
                for j in range(i))
    print("n =", n, "order =", order)
    print(A)
    print("upper triangular:", upper)
    print("determinant:", A.det())
```

The method `m.transition_matrix(s,n)` writes the degree- n monomial basis in the Schur basis. Sage orders partitions in decreasing lexicographic order. The matrices are

n=1: [1]

n=2: [1 -1]	n=3: [1 -1 1]
[0 1]	[0 1 -2]
	[0 0 1]

n=4:
[1 -1 0 1 -1]
[0 1 -1 -1 2]

```
[0  0  1 -1  1]
[0  0  0  1 -3]
[0  0  0  0  1]
```

n=5:

```
[1 -1  0  1  0 -1  1]
[0  1 -1 -1  1  1 -2]
[0  0  1 -1 -1  2 -2]
[0  0  0  1 -1 -1  3]
[0  0  0  0  1 -2  3]
[0  0  0  0  0  1 -4]
[0  0  0  0  0  0  1]
```

n=6:

```
[1 -1  0  1  0  0 -1  0  0  1 -1]
[0  1 -1 -1  0  1  1  0 -1 -1  2]
[0  0  1 -1 -1  0  1 -1  1 -2  2]
[0  0  0  1  0 -1 -1  1  1  1 -3]
[0  0  0  0  1 -1  1  1  0 -1  1]
[0  0  0  0  0  1 -2 -2  0  4 -6]
[0  0  0  0  0  0  1  0 -1 -1  4]
[0  0  0  0  0  0  0  1 -1  1 -1]
[0  0  0  0  0  0  0  0  1 -3  6]
[0  0  0  0  0  0  0  0  0  1 -5]
[0  0  0  0  0  0  0  0  0  0  1]
```

Every matrix is upper triangular with diagonal entries 1, so every determinant is 1. This is the computational form of the unitriangularity of the Kostka matrix.

2. **Solution.** Here are examples with fewer than four parts, exactly four parts, and more than four parts.

```
examples = [[2], [2,1], [1,1,1,1], [1,1,1,1,1]]
for la in examples:
    print(la, s[la].expand(4))
    print("monomial form:", m(s[la]))
```

Sage uses variables x_0, x_1, x_2, x_3 . The mathematical results are

$$s_2 = m_2 + m_{1,1}, \quad s_{2,1} = m_{2,1} + 2m_{1,1,1}, \quad s_{1^4} = m_{1^4}.$$

The abstract function $s_{1^5} = m_{1^5}$ is nonzero. In contrast, the command

```
s[1,1,1,1,1].expand(4)
```

returns 0, because a Schur polynomial whose shape has more than four rows vanishes in four variables.

Thus `expand(4)` realizes an abstract symmetric function as a symmetric polynomial in four variables. The monomial coefficients are Kostka numbers: they are nonnegative integers counting semistandard tableaux.

3. **Solution.** The following loop performs every requested calculation.

```
families = [
    ("paths",    range(1,8), graphs.PathGraph),
    ("cycles",   range(3,8), graphs.CycleGraph),
    ("complete", range(1,8), graphs.CompleteGraph)
]
bases = [("m",m), ("e",e), ("p",p), ("s",s)]

for name, sizes, constructor in families:
    print("\n", name)
    for n in sizes:
        G = constructor(n)
        X = G.chromatic_symmetric_function(QQ)
        print("n =", n)
        for basis_name, B in bases:
            print(basis_name, B(X))
```

`chromatic_symmetric_function()` first returns the power-sum expansion, using

$$X_G = \sum_{A \subseteq E(G)} (-1)^{|A|} p_{\lambda(A)}.$$

The numbers of nonzero terms in the four bases are:

family and sizes	basis	numbers of terms
$P_n, 1 \leq n \leq 7$	m, s	(1, 1, 2, 3, 5, 7, 11)
	e	(1, 1, 2, 3, 4, 6, 8)
	p	(1, 2, 3, 5, 7, 11, 15)
$C_n, 3 \leq n \leq 7$	m, s	(1, 3, 3, 7, 8)
	e	(1, 2, 2, 4, 4)
	p	(3, 5, 7, 11, 15)
$K_n, 1 \leq n \leq 7$	m, e, s	(1, 1, 1, 1, 1, 1, 1)
	p	(1, 2, 3, 5, 7, 11, 15)

For paths and cycles in this range, the elementary expansion has the fewest terms. For complete graphs,

$$X_{K_n} = n!e_n,$$

so the elementary basis is especially natural. All three families have nonnegative coefficients in the monomial, elementary, and Schur bases for these values of n . Negative power-sum coefficients occur as soon as $n \geq 2$.

4. **Solution.** Sage's `StarGraph(r)` has one center and r leaves, so it has $r + 1$ vertices.

```
for n in range(1, 8):
    St = graphs.StarGraph(n-1)
    X = p(St.chromatic_symmetric_function(QQ))
    formula = sum((-1)^r * binomial(n-1,r)
                  * p[Partition([r+1] + [1]*(n-1-r))])
    for r in range(n):
        print(n, X, X == formula)
```

Every comparison returns True. Thus

$$X_{St_n} = \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} p_{r+1, 1^{n-1-r}}.$$

This also follows directly from the edge-subset formula: choosing r of the $n - 1$ star edges creates one component of size $r + 1$ and $n - 1 - r$ isolated vertices.

5. **Solution.**

```
from collections import defaultdict
```

```

for n in [7,8]:
    groups = defaultdict(list)
    trees = list(graphs.trees(n))
    for i,T in enumerate(trees, start=1):
        X = T.chromatic_symmetric_function(QQ)
        groups[X].append(i)
    collisions = [I for I in groups.values() if len(I) > 1]
    print(n, len(trees), collisions)

```

Sage generates one representative from each isomorphism class. It finds 11 trees for $n = 7$ and 23 trees for $n = 8$. In both cases `collisions` is empty. Thus no two trees in either collection have the same chromatic symmetric function.

6. **Solution.** The code below computes the elementary expansion and groups trees by the requested coefficient.

```

trees8 = list(graphs.trees(8))
groups = defaultdict(list)

for i,T in enumerate(trees8, start=1):
    XE = e(T.chromatic_symmetric_function(QQ))
    c = XE[Partition([4,2,1,1])]
    groups[c].append(i)
    print(i, XE, "coefficient =", c)

print({c:I for c,I in groups.items() if len(I) > 1})

```

With trees numbered in Sage's iterator order, the repeated groups are

coefficient	tree indices
0	1, 16
3	4, 10
6	2, 7
10	3, 14
12	9, 22
13	8, 18

Therefore the coefficient of $e_{4,2,1,1}$ does not distinguish all trees on eight vertices.

7. **Solution.** This program computes all five requested statistics.

```
for i,T in enumerate(trees8, start=1):
    X = T.chromatic_symmetric_function(QQ)
    XE, XP = e(X), p(X)
    deg = T.degree_sequence()
    leaves = deg.count(1)
    print(i,
          "degree sequence =", deg,
          "diameter =", T.diameter(),
          "leaves =", leaves,
          "e[7,1] =", XE[Partition([7,1])],
          "p[7,1] =", XP[Partition([7,1])])
```

The coefficient pair depends only on the number L of leaves:

L	2	3	4	5	6	7
$[e_{7,1}]X_T$	6	13	20	27	34	41
$[p_{7,1}]X_T$	2	3	4	5	6	7

In fact, for a tree on n vertices,

$$[p_{n-1,1}]X_T = (-1)^{n-2}L, \quad [e_{n-1,1}]X_T = (n-1)L - n.$$

For the power coefficient, a contributing edge subset must isolate one leaf and contain all edges of the remaining $(n-1)$ -vertex tree. Thus these coefficients recover the number of leaves, which is already visible in the degree sequence. They do not determine the diameter: different trees with the same number of leaves can have different diameters.

Mini-Projects

A. **Solution.** The following code produces a reproducible data record for every tree.

```
def basis_statistics(f):
    coeffs = f.coefficients()
    return {
        "terms": len(f.support()),
        "smallest coefficient": min(coeffs),
```

```

        "largest coefficient": max(coeffs),
        "all positive": all(c > 0 for c in coeffs)
    }

records = []
for i,T in enumerate(trees8, start=1):
    X = T.chromatic_symmetric_function(QQ)
    row = {"tree": i, "graph6": T.graph6_string()}
    for name,B in [("m",m),("e",e),("p",p),("s",s)]:
        row[name] = basis_statistics(B(X))
    records.append(row)
    print(row)

```

The total numbers of nonzero terms over all 23 trees are

basis	<i>m</i>	<i>e</i>	<i>p</i>	<i>s</i>
total terms	384	377	416	401
average	16.70	16.39	18.09	17.43

There is no basis that is sparsest for every tree. Counting ties, the elementary basis is sparsest for 15 trees, the monomial and power bases for 7 each, and the Schur basis for 3.

All 23 monomial expansions have positive coefficients, as follows directly from the coloring definition. Only one of these trees (the path) has all positive elementary coefficients in this experiment, and 9 have all positive Schur coefficients. None has all positive power-sum coefficients. The coefficients also encode graph data: for example, Exercise 7 shows how to recover the number of leaves.

B. Solution. Sage partitions can be sorted directly in lexicographic order.

```

for i,T in enumerate(trees8, start=1):
    X = T.chromatic_symmetric_function(QQ)
    print("tree", i)
    for name,B in [("m",m),("e",e),("p",p),("s",s)]:
        support = sorted(B(X).support())
        print(name, "smallest =", support[0],
              "largest =", support[-1])

```

In the monomial basis, the smallest partition is always 1^8 . The first part of the largest partition is the independence number $\alpha(T)$: a color class is an independent set, and lexicographic maximization first makes the largest color class as large as possible. Thus the largest monomial index contains genuine information about the tree. In the power basis, connectedness forces a nonzero p_8 term, so the largest index is always (8) and is not useful for distinguishing these trees by itself.

To study the effect of adding a leaf, attach it at every possible vertex and recompute the dictionaries of coefficients:

```
def add_leaf(T, v):
    U = T.copy()
    new_vertex = max(U.vertices(sort=False)) + 1
    U.add_edge(v, new_vertex)
    return U

for i,T in enumerate(trees8, start=1):
    for v in T.vertices(sort=False):
        U = add_leaf(T, v)
        XU = U.chromatic_symmetric_function(QQ)
        print("tree", i, "attachment vertex", v)
        for name,B in [("m",m),("e",e),("p",p),("s",s)]:
            print(name, B(XU).monomial_coefficients())
```

There is no single coefficient change independent of the attachment vertex. Attaching at different vertices can produce nonisomorphic trees and different expansions. The code makes that dependence explicit and repeats the experiment in all four bases.